MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

The radius of ${}_{72}\mathrm{Te}^{125}$ nucleus is 6 fermi. The radius of ${}_{13}\mathrm{Al}^{27}$ nucleus in meters is

(A) 3.6×10^{-12} m

(B) 3.6×10^{-15} m

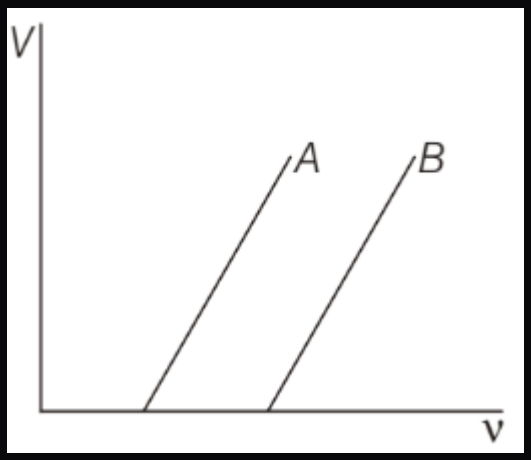
(C) 7.2×10^{-8} m

(D) 7.2×10^{-15} m

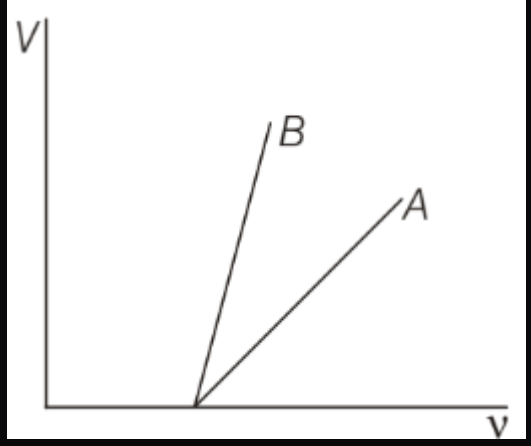
Q.2

When light of frequency v incidents on two metallic plates A and B, photo electrons are emitted. If the work function of A is more than that of B, the correct curve of the following curves drawn between stopping potential V and incident frequency v is

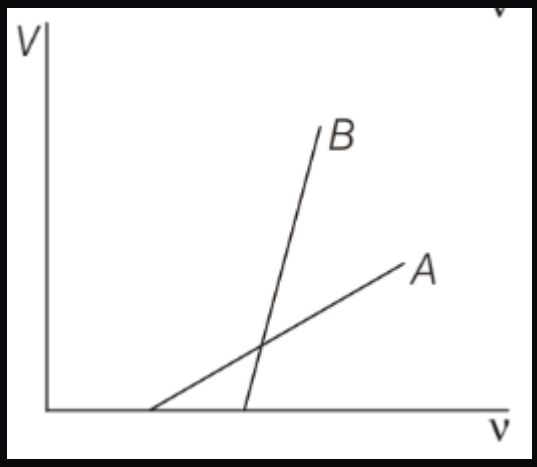
(A)



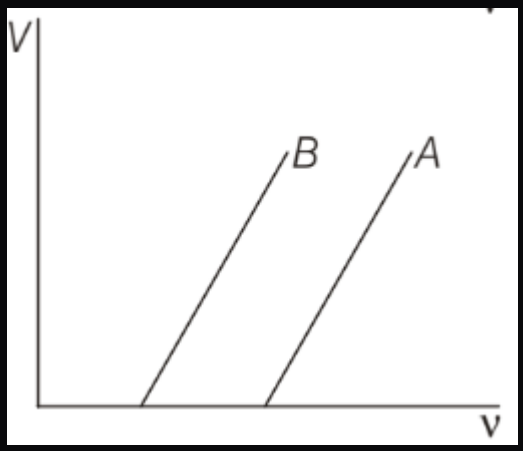
(B)



(C)

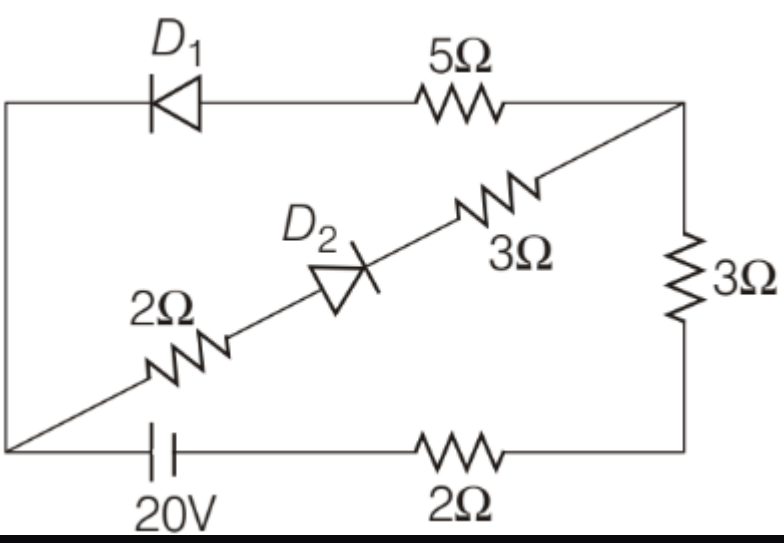


(D)



Q.3

If the diodes are ideal in the circuit given below, then the current through the cell is



(A) 4 A

(B) 1.5 A

(C) 2 A

(D) 3 A

Q.4

Unpolarised light from air incidents on the surface of a transparent medium of refractive index 1.414 such that the reflected light is completely polarised. Match the angles given in List-I with the corresponding values given in List-II.

List-I		List-II
A. Angle of reflection	(i)	$2\sin^{-1}\left(\sqrt{\frac{2}{3}}\right)$
B. Angle of refraction	(ii)	$\sin^{-1}\left(\sqrt{\frac{2}{3}}\right) - \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$
C. Angle between incident and completely polarised light	(iii)	$\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$
D. Angle of deviation of the incident ray	(iv)	$\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

The correct match is

Codes

A *B* *C* *D*

(A) (ii) (iii) (i) (iv)

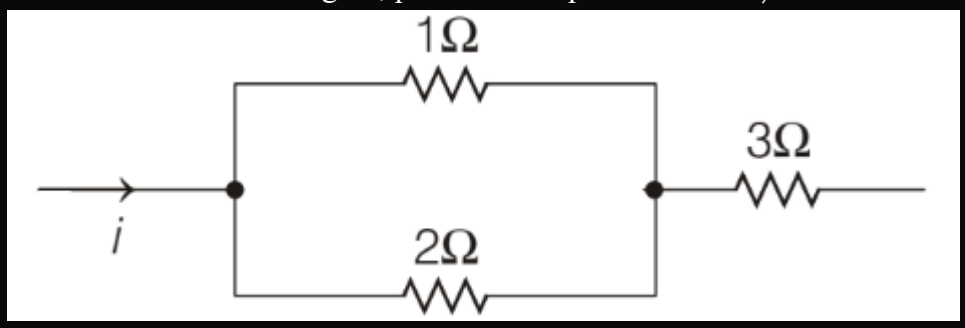
(B) (ii) (iii) (iv) (i)

(C) (iv) (i) (iii) (ii)

(D) (iv) (iii) (i) (ii)

Q.5

In the circuit shown in figure, power developed across $1\Omega, 2\Omega$ and 3Ω resistances are in the ratio.



(A) 1 : 2 : 3

(B) 4 : 2 : 27

(C) 6 : 4 : 9

(D) 2 : 1 : 27

Q.6

The minimum force required to move a body up an inclined plane is three times the minimum force required to prevent it from sliding down the plane. If the coefficient of friction between the body and the inclined plane is $\frac{1}{2\sqrt{3}}$, then the angle of the inclined plane is

(A) 60°

(B) 45°

(C) 30°

(D) 15°

Q.7

The escape velocity of a body on the earth's surface is v_e.A body is thrown up with a speed $\sqrt{5}v_e$.Assuming that the sun and planets do not influence the motion of the body, velocity of the body at infinite distance, is :

(A) 0

(B) v_e

(C) $\sqrt{2}v_e$

(D) $2v_e$

Q.8

Speed of electron in its Ist Bohr's orbit is given by $2.18 \times 10^6\ \mathrm{ms}^{-1}$. If the time period of electron in nth orbit is measured as 4.10 femto second, the value of n is

(A) 1

(B) 2

(C) 3

(D) 4

Q.9

Why $220\ \mathrm{V}$ AC is more dangerous than $220\ \mathrm{V}$ DC?

(A) The DC attracts

(B) Peak voltage for AC is much larger

(C) The body offers less resistance to AC

(D) Due to some other reason

Q.10

Which of the following statement is correct regarding $p - V$ graph?
1. Slope of $p - V$ graph in an isothermal process is $-\frac{p}{V}$.
2. Slope of $p - V$ graph in an adiabatic process is $-\frac{p}{V}$.
3. Slope of $p - V$ graph in an isochoric process is $-\frac{\gamma p}{V}$.
4. Slope of $p - V$ graph in an isobaric process is zero.

(A) 1, 3, 4 are correct

(B) 2, 3 are correct

(C) 1, 4 are correct

(D) 2,3,4 are correct

Q.11

The equations of motion of a projectile are given by $x = 36t$ metre and $2y = 96t - 9.8t^2$ metre. The angle of projection is :

(A) $\sin^{-1}\left(\frac{4}{5}\right)$

(B) $\sin^{-1}\left(\frac{3}{5}\right)$

(C) $\sin^{-1}\left(\frac{4}{3}\right)$

(D) $\sin^{-1}\left(\frac{3}{4}\right)$

Q.12

If A represents Boltzmann constant, B represents Planck's constant and C represents speed of light in vacuum, then the quantity having the dimensions of $A^4B^{-3}C^{-2}$ is

(A) universal gas constant

(B) specific heat capacity

(C) stefan's constant

(D) heat energy

Q.13

The specific heat of helium at constant volume is 12.6 $\mathrm{Jmol^{-1}\ K^{-1}}$. The specific heat of helium at constant pressure in $\mathrm{Jmol^{-1}\ K^{-1}}$ is about (Assume the temperature of the gas is moderate, universal gas constant, $R = 8.314\mathrm{Jmol^{-1}\ K^{-1}}$)

(A) 12.6

(B) 16.8

(C) 18.9

(D) 21

Q.14

A bomb at rest explodes into three parts of equal mass. If the momentum of two parts are $-2p\hat{\mathbf{i}}$ and $p\hat{\mathbf{j}}$, find the magnitude of momentum of the third part.

(A) p

(B) $\sqrt{3}p$

(C) $\sqrt{5}p$

(D) 0

Q.15

A mass M kg is suspended by a weightless string. The horizontal force required to hold the mass at 60° with the vertical is

(A) Mg

(B) $Mg\sqrt{3}$

(C) $Mg(\sqrt{3}+1)$

(D) $\frac{Mg}{\sqrt{3}}$

Q.16

For a screw gauge least count of the main scale is 1 mm. Find the minimum number of divisions on the circular scale so that it can measure a diameter of 5 μm

(A) 500

(B) 100

(C) 50

(D) 200

Q.17

A system consists of two springs connected in series and each having the spring constant $10\mathrm{Nm}^{-1}$. The minimum work required to stretch this system by lcm in erg is

(A) 1500

(B) 2000

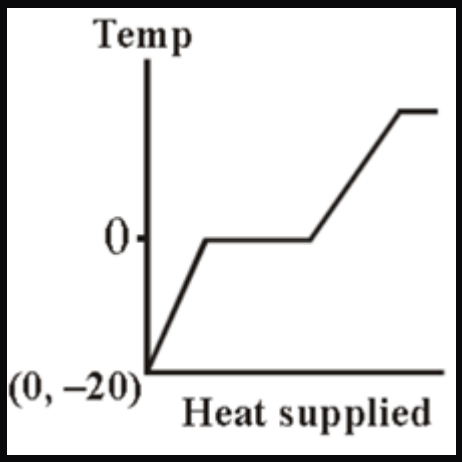
(C) 3000

(D) 2500

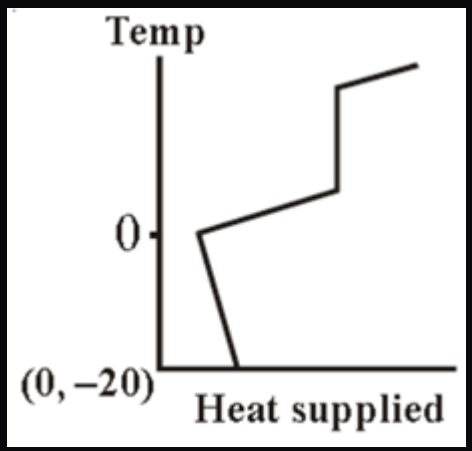
Q.18

A block of ice at temperature $-20^{\circ}\mathrm{C}$ is slowly heated and converted to steam at $100^{\circ}\mathrm{C}$. Which of the following diagram is most appropriate?

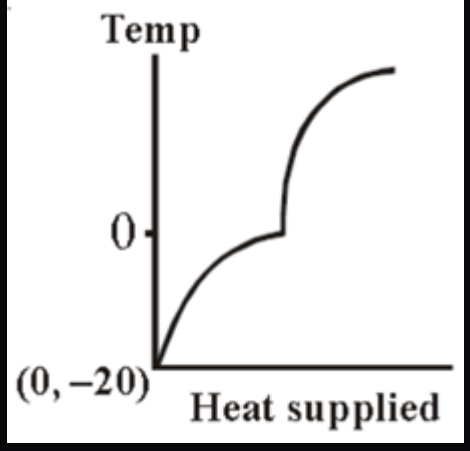
(A)



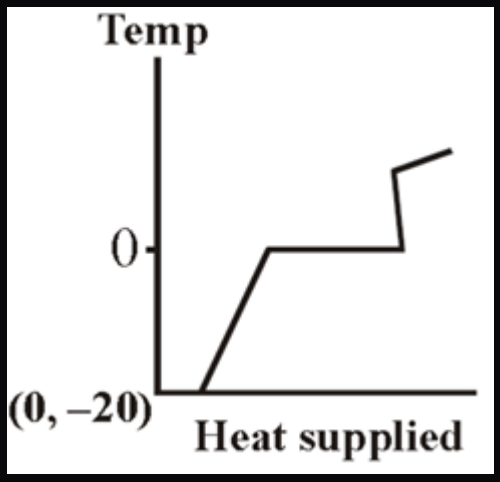
(B)



(C)



(D)



Q.19

Two point charges $-q$ and $+q$ are located at points $(0, 0, -a)$ and $(0, 0, a)$ respectively. The electric potential at a point $(0, 0, z)$, where $z > a$ is

(A) $\frac{qa}{4\pi\varepsilon_0 z^2}$

(B) $\frac{q}{4\pi\varepsilon_0 a}$

(C) $\frac{2qa}{4\pi\varepsilon_0(z^2-a^2)}$

(D) $\frac{2qa}{4\pi\varepsilon_0(z^2+a^2)}$

Q.20

The correct ascending order of the wavelengths of the following electromagnetic waves is
Ultraviolet, X-rays, infrared, gamma rays and visible light.

(A) X-ray, gamma ray, ultraviolet, infrared, visible light.

(B) Gamma ray, X-ray, ultraviolet, visible light, infrared.

(C) Gamma ray, X-ray, infrared, visible light, ultraviolet:

(D) Gamma ray, X-ray, ultraviolet, infrared, visible light.

Q.21

The intensity of light emerging from one of the slits in a Young's double slit experiment is found to be 1.5 times the

intensity of light emerging from the other slit. What will be the approximate ratio of intensity of an interference maximum to that of an interference minimum

Q.22

The number of significant figures in quantity $0.00005041\ \mathrm{J}$ is

Q.23

When a glass prism of refracting angle 60° is immersed in a liquid its angle of minimum deviation is 30°. The critical angle of glass with respect to the liquid medium is (Mark answer in degree)

Q.24

Two cells P and Q each of emf $2.16\ \mathrm{V}$ are connected in series with a resistor of 19.6Ω. An ideal voltmeter reads $2\ \mathrm{V}$, when connected across the cell P and $1.92\ \mathrm{V}$ when connected across the cell Q. The ratio of the internal resistances of the cell P and Q is K/6, find K.

Q.25

A region contains a uniform electric field $\mathbf{E} = (10\hat{\mathbf{i}} + 30\hat{\mathbf{j}})\mathrm{Vm}^{-1}$. A and B are two points in the field at $(1, 2, 0)\mathrm{m}$ and $(2, 1, 3)\mathrm{m}$, respectively. The work done when a charge of $0.8\mathrm{C}$ moves from A to B in a parabolic path is

Q.26

Observe the following statements :
I. The physical and chemical properties of elements are periodic functions of their electronic configuration.
II. Electronegativity of fluorine is less than the electronegativity of chlorine.
III. Electropositive nature decreases from top to bottom in a group.
The correct answer is :

(A) I, II and III are correct

(B) only I is correct

(C) only I and II is correct

(D) only II and III are correct

Q.27

Which of the following compounds is expected to be coloured?

(A) $\mathrm{ZnSO_4}$

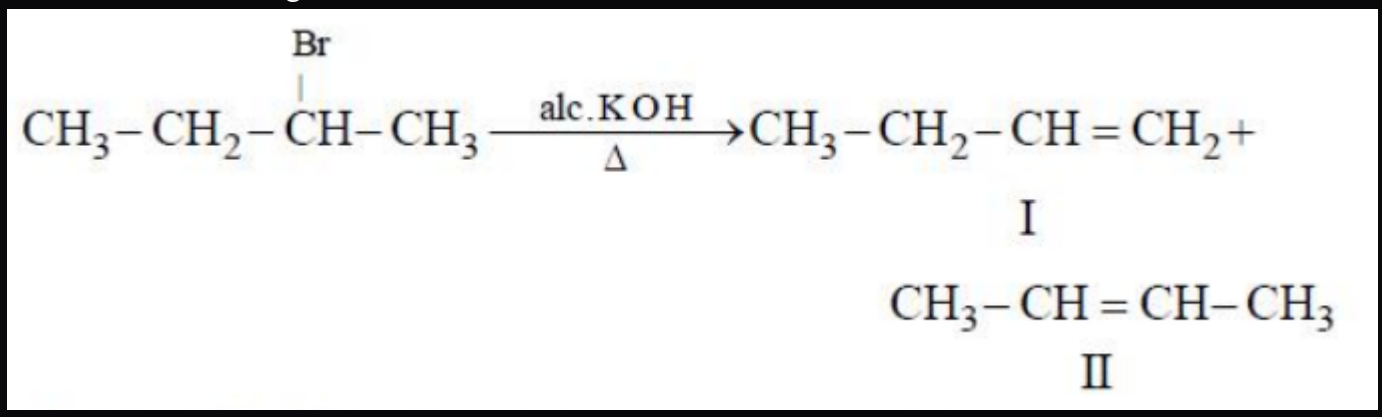
(B) $\mathrm{MgCl_2}$

(C) $\mathrm{CuCl_2}$

(D) AgCl

Q.28

Consider the following reaction



Which of the following statements are correct ?

a. I is the major product of the reaction

b. II is the major product of the reaction

c. Formation of I is in accordance with Saytzeff's rule

d. II is more stable because it is more substituted

(A) a,c

(B) b,c

(C) a,d

(D) b,d

Q.29

One mole of alkene $\underline{X}$ on ozonolysis gave one mole of acetaldehyde and one mole of acetone. The IUPAC name of $\underline{X}$ is

(A) 2-methyl-2-butene

(B) 2-methyl-1-butene

(C) 2-butene

(D) 1-butene

Q.30

2 g of a non-electrolyte solute (molar mass is $500\ \text{g mol}^{-1}$) was dissolved in 57.3 g of xylene. If the freezing point depression constant K_f of xylene is $4.3\ \text{K kg mol}^{-1}$. Then, the depression in freezing point of xylene is.....

(A) 57.3 K

(B) 0.3 K

(C) 4.3 K

(D) 0.002 K

Q.31

The electron affinity values of elements A, B, C and D are respectively $-135, -60, -200$ and $-348\ \mathrm{kJ\ mol\ m^{-1}}$. The outer electronic configuration of element B is

(A) $3s^2 3p^5$

(B) $3s^2 3p^4$

(C) $3s^2 3p^3$

(D) $3s^2 3p^2$

Q.32

x grams of water is mixed in $69\ \mathrm{g}$ of ethanol. Mole fraction of ethanol in the resultant solution is 0.6 . What is the value of x in grams?

(A) 54

(B) 36

(C) 180

(D) 18

Q.33

Which one of the following has highest dipole moment?

(A) cis-but-2-ene

(B) trans-1, 2-dichloroethene

(C) cis-1, 2-dichloroethene

(D) trans-but-2-ene

Q.34

Which of the following belongs to the homologous series of $\mathrm{C_5H_8O_2\ N}$?

(A) $\mathrm{C_6H_{10}O_3\ N}$

(B) $\mathrm{C_6H_8O_2\ N_2}$

(C) $\mathrm{C_6H_{10}O_2\ N_2}$

(D) $\mathrm{C_6H_{10}O_2\ N}$

Q.35

On treating $\mathrm{SO_2}$ with aqueous solution of $\mathrm{KMnO_4}$, the manganese ion reduces to

(A) $\mathrm{Mn^{2+}}$ only

(B) $\mathrm{Mn^{4+}}$ only

(C) Mn^{6+} only

(D) Mn^{4+} and Mn^{6+}

Q.36

Which complex among the following has the highest value of spin only magnetic moment?
$[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$, $[Ni(CN)_4]^{2-}$, $[NiCl_4]^{2-}$

(A) $[Fe(CN)_6]^{3-}$

(B) $[Fe(CN)_6]^{4-}$

(C) $[Ni(CN)_4]^{2-}$

(D) $[NiCl_4]^{2-}$

Q.37

Assertion (A): VF_5 is stable while VCl_5 is not.
Reason (R) : Fluorine stabilises the higher oxidation state due to its higher bond enthalpy.
The correct option among the following is

(A) (A) is true, (R) is true and (R) is the correct explanation for (A)

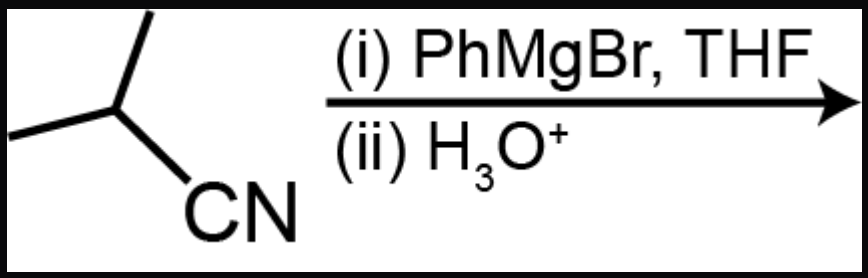
(B) (A) is true, (R) is true but (R) is not the correct explanation for (A)

(C) (A) is true but (R) is false

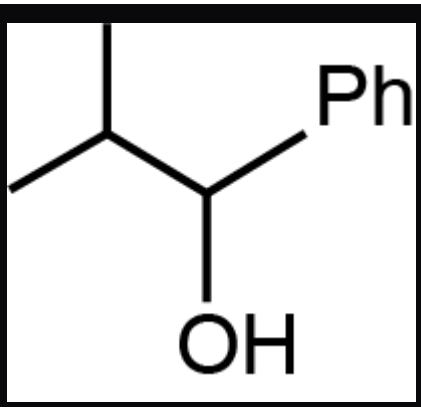
(D) (A) is false but (R) is true

Q.38

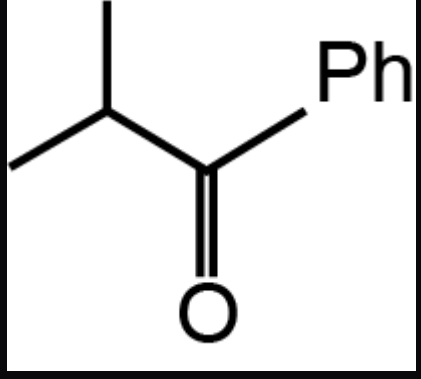
For the reaction below the structure of the product Q is



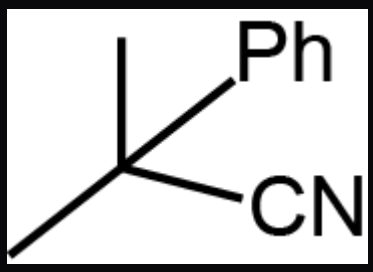
(A)



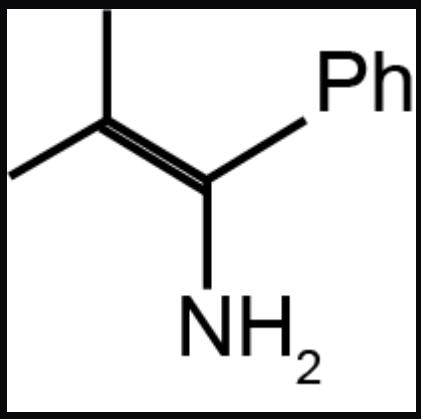
(B)



(C)



(D)



Q.39

In an organic compound phosphorus is estimated as

(A) $Mg_3(PO_4)_2$

(B) P_2O_5

(C) $Mg_2P_2O_7$

(D) H_3PO_4

Q.40

Which of the following statements is correct for the cell $Zn\left|Zn^{+2}\right||Cu^{+2}\left|Cu\right.$?

(A) Zn^{+2} is reducing agent.

(B) Cu is anode.

(C) Cu is oxidising agent.

(D) The cell reaction is $Zn + Cu^{+2} \longrightarrow Zn^{+2} + Cu$

Q.41

Amongst the following compounds, the one that will not respond to Cannizzaro reaction upon treatment with alkali is

(A) Cl_3CCHO

(B) Me_3CCHO

(C) C_6H_5CHO

(D) HCHO

Q.42

Which intermediate is formed in the Reimer-Tiemann reaction?

(A) Aldehyde

(B) Carbocation

(C) Carbanion

(D) Substituted benzal chloride

Q.43

Ferrous ion charges to X ion, on reacting with acidified hydrogen peroxide.
The number of d-electrons present in X and its magnetic moment (in BM) are, respectively :

(A) 6 and 6.93

(B) 5 and 5.92

(C) 5 and 4.9

(D) 4 and 5.92

Q.44

The correct order for increasing reactivity towards S_N1 substitution for the following compounds is

1. CH_3CH_2Br 2. $CH_2 = CH - CH(Br)CH_3$
3. $CH_2 = CHBr$ 4. $CH_3 - CH(Br)CH_3$

(A) $1 < 4 < 3 < 2$

(B) $4 < 1 < 2 < 3$

(C) $3 < 1 < 2 < 4$

(D) $3 < 1 < 4 < 2$

Q.45

Identify, from the following, the diamagnetic, tetrahedral complex

(A) $[Ni(Cl)_4]^{2-}$

(B) $[Co\,(C_2O_4)_3]^{3-}$

(C) $[Ni(CN)_4]^{2-}$

(D) $[Ni(CO)_4]$

Q.46

When $BaCl_2$ is added to an aqueous salt solution, a white precipitate is obtained. How many anion among CO_3^{2-}, SO_3^{2-} and SO_4^{2-} can be present in the solution

Q.47

One mole of N_2H_4 loses 10 moles of electrons to form a new compound Z. Assuming that all the nitrogens appear in the new compound, what is the oxidation state of nitrogen in Z ? (There is no change in the oxidation state of hydrogen.)

Q.48

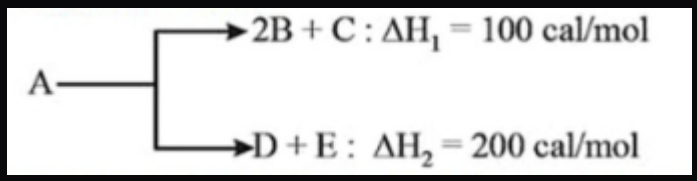
An element has d^5 configuration. The total number of electron exchanges possible for it is

Q.49

The amount of sucrose needed to produce 1 mole of glucose using acid hydrolysis is (mark answer in grams)

Q.50

For the reaction :



If initially $1\ mol$ of A is taken & finally ratio of B & D is $6:1$ what is total enthalpy change (cal) per mole of A reacted.

Q.51

Coefficient of x^5 in $\frac{(1-x^2)^3(1-x^3)(1-x^4)}{(1-x)^5}$ is -

(A) 16

(B) 19

(C) 20

(D) 15

Q.52

It is given that $\log_6 a + \log_6 b + \log_6 c = 6$, where a, b and c are positive integers that form an increasing geometric sequence and $b - a$ is the square of an integer. Find $a + b + c$.

(A) 120

(B) 111

(C) 132

(D) 96

Q.53

A straight line through $\mathrm{A}(6,8)$ meets the curve $2\mathrm{x}^2+\mathrm{y}^2=2$ at B and C. P is a point on BC such that $\mathrm{AB}, \mathrm{AP}, \mathrm{AC}$ are in H.P, then the minimum distance of the origin from the locus of ' P ' is

(A) $\frac{1}{\sqrt{52}}$

(B) $\frac{5}{\sqrt{52}}$

(C) $\frac{10}{\sqrt{52}}$

(D) $\frac{15}{\sqrt{52}}$

Q.54

Length of latus rectum of the parabola $9x^2+16y^2+24xy-4x+3y=0$ is

(A) $\frac{1}{20}$

(B) $\frac{1}{4}$

(C) $\frac{1}{5}$

(D) 1

Q.55

Let A be matrix $\left[\begin{array}{cc}1 & 1 \\ -1 & 1\end{array}\right]$ then the matrix $B=A^{10}-A^9+2A^8-A^7+4A^6-2A^5+4A^4+A^3-A^2+A+$ I is

(A) A

(B) $A+I$

(C) 1

(D) $\mathrm{A}-\mathrm{I}$

Q.56

The straight line passing through the point $(8,4)$ and cuts y-axis at Bx-axis at A. The locus of mid point of AB is

(A) $xy+2x+4y=64$

(B) $xy-2x-4y=0$

(C) $xy-4x-2y+8=0$

(D) $xy+4x+2y=72$

Q.57

Let $\vec{a}=2\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+3\hat{j}-5\hat{k}$ be two vectors, and $\overrightarrow{\mathrm{r}}$ be a vector along the vector $3\overrightarrow{\mathrm{a}}-2\overrightarrow{\mathrm{b}}$ such that $|\overrightarrow{\mathrm{r}}|=\sqrt{74}$. If the direction of $\vec{r}$ is opposite to that of $3\vec{a}-2\vec{b}$, then $\overrightarrow{\mathrm{r}}=$

(A) $-7\hat{\mathrm{i}}-4\hat{\mathrm{j}}+3\hat{\mathrm{k}}$

(B) $4\hat{i}+7\hat{j}-3\hat{k}$

(C) $-4\hat{\mathrm{i}}+3\hat{\mathrm{j}}-7\hat{\mathrm{k}}$

(D) $4\hat{\mathrm{i}}-3\hat{\mathrm{j}}+7\hat{\mathrm{k}}$

Q.58

Consider two sets

$\mathrm{A}=\left\{\frac{\mathrm{p}}{\mathrm{q}};\mathrm{p},\mathrm{q}\right.$ are coprime odd integers $\}$ and

$B=\left\{\frac{p+1}{q}:p,q\right.$ are odd integers and $p\neq-1\}$.

Functions $f:\mathrm{A}\to\mathrm{R}$ & $\mathrm{g}:\mathrm{B}\to\mathrm{R}$ are both defined as $f(\mathrm{x})=\mathrm{x}^{\mathrm{x}}\&\ \mathrm{g}(\mathrm{x})=\mathrm{x}^{|\mathrm{x}|}$ on respective domains. Then mark incorrect option

(A) $f\left(\frac{1}{3}\right)\cdot g\left(\frac{-2}{3}\right)=\frac{\sqrt[3]{4}}{3}$

(B) $g(x)$ is an even function.

(C) $f(1)+g(2)=6$

(D) $\left[g\left(\frac{-6}{5}\right)\right]^{5/2}-\left[g\left(\frac{4}{5}\right)\right]^{15/4}-\left[f\left(\frac{3}{5}\right)\right]^{5}=1$

Q.59

In a tetrahedron OABC, the edges are of lengths, $|\mathrm{OA}|=|\mathrm{BC}|=\mathrm{a},|\mathrm{OB}|=|\mathrm{AC}|=\mathrm{b},|\mathrm{OC}|=|\mathrm{AB}|=$ c. Let G_1 and G_2 be the centroids of the triangle ABC and AOC such that $\mathrm{OG}_1\perp\mathrm{BG}_2$, then the value of $\frac{a^2+c^2}{b^2}$ is (O being origin)

(A) 2

(B) 3

(C) 6

(D) 9

Q.60

Let $P(x)=a_0+a_1x^2+a_2x^4+\ldots+a_nx^{2n}$ with all $a_i>0,i=0,1,2,\ldots,n$. Then $P(x)$ has

(A) neither maximum nor minimum

(B) only one maximum

(C) only one minimum

(D) one maximum and one minimum

Q.61

If $f(x)=\begin{cases}\left(p^2-1\right)(\{x\}+2[x])-2, & -2<x\leq -1\\ q\left(\frac{e^x+e^{-x}}{2}\right)+|p|(x-1), & -1<x<2\end{cases}$, $p,q\in R$ is continuous in $(-2,2)$ then $f\left(f\left(f\left(\frac{-1}{2}\right)\right)\right)$ is

[Note: $[\mathrm{k}]$ denotes greatest integer function less than or equal to k and $\{k\}$ denotes fractional part function of k.]

(A) -2

(B) -1

(C) 0

(D) not defined

Q.62

The sum of maximum & minimum values of expression $y=10\sin^2 x+6\sin x\cos x+14\cos^2 x$ is-

(A) 12

(B) 24

(C) 36

(D) 48

Q.63

4 cards are drawn from a well shuffled pack of 52 cards without replacement. If the fourth card is a king, then the chance that the first 3 cards are also king is $\frac{1}{20820+a}$, then a is equal to

(A) 4

(B) 1

(C) 2

(D) 5

Q.64

Consider a real valued continuous function f such that $f(x)=\sin x+\int_{-2}^{x/2}(\sin x+tf(t))dt$ then the difference of maximum, minimum value of f in $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$ is

(A) $\pi+1$

(B) $2(\pi+1)$

(C) π

(D) can't be determined

Q.65

If $x^2-x+1=0$, then $\sum_{r=1}^{2011}\left(x^r+\frac{1}{x^r}\right)^3=$

(A) 0

(B) 1

(C) -1

(D) none of these

Q.66

If $f(x)=\frac{ax+b}{bx-a}, x \neq \frac{a}{b}, a>0, b>0$ and if $g(x)=f(f(x))+f\left(f\left(\frac{1}{x}\right)\right)$ and $g:[1, \infty) \rightarrow[2, \infty)$ then the value of x for which $g(x)=\frac{5}{2}$, is

(A) 4

(B) 3.5

(C) 2

(D) 4.5

Q.67

Let A and B be two 3×3 invertible matrices such that A is an idempotent matrix and $\det(\operatorname{adj} B)=\det(A)^{12}+\det\left(A-A^{\top}\right)^{13}+\det(\operatorname{adj} A-I)^{14}+L$
where $L=\lim_{x \rightarrow 0} \frac{\int_0^{\sin^2 x} \ln\left(1+t^3\right) dt}{\int_0^x\left(e^t-1-t\right)(\tan t-t)(1-\cos t) dt}$
Then the value of $\det(B)$

(A) 4

(B) 3

(C) 2

(D) 5

Q.68

The function $\mathrm{f(x)=\max\{3-x, 3+x, 6\}}$ is differentiable at

(A) All points

(B) No point

(C) All points except two points

(D) All points expect at one point

Q.69

Consider $\mathrm{I_1=\int_0^{\pi/2}(\cos x)^{\sqrt{17}+3} dx}$ and $\mathrm{I_2=\int_0^{\pi/2}(\cos x)^{\sqrt{17}+5} dx}$, then $\frac{\mathrm{I_1}}{\mathrm{I_2}}$ is equal to-

(A) $\frac{\sqrt{17}+3}{\sqrt{17}+5}$

(B) $\frac{8}{\sqrt{17}+3}$

(C) $\sqrt{17}+3$

(D) None of these

Q.70

A line passing through a point $(2,-1,3)$ and paraliel to the line $\overrightarrow{\mathrm{r}}=\hat{\mathrm{i}}-\hat{\mathrm{j}}+\hat{\mathrm{k}}+\lambda(\hat{\mathrm{i}}+\hat{\mathrm{j}}+\hat{\mathrm{k}})$ meets xy, yz and zx planes in points A, B and C respectively, then centroid of the triangle ABC is -

(A) $\left(\frac{2}{3}, \frac{7}{3}, \frac{5}{3}\right)$

(B) $\left(\frac{2}{3}, -\frac{7}{3}, \frac{5}{3}\right)$

(C) $\left(-\frac{2}{3}, -\frac{7}{3}, \frac{5}{3}\right)$

(D) $\left(-\frac{2}{3}, -\frac{7}{3}, -\frac{5}{3}\right)$

Q.71

Area bounded between the curves given by $|x|+y=7$ and $4y=\left|4-x^2\right|$, is (in sq. units)

Q.72

Three circles each of radius 3 are drawn with centers at $(0,16),(3,0)$ and $(5,8)$. A line of slope m passing through $(3,0)$ is such that the total area of the part of the three circles to one side of the line is equal to the total area of the part of the three circles to the other side of it. Then $\left|\frac{\mathrm{m}}{8}\right|$ is equal to

Q.73

The variance of the following continuous frequency distribution is

Class Interval	$0-10$	$10-20$	$20-30$	$30-40$
Frequency	2	3	4	1

Q.74

Let $\sum_{k=0}^{1000}\left(\begin{array}{c}2000\\2k\end{array}\right)9^k$ is divisible by 2^a, then maximum value of a is

Q.75

Let $a_1, a_2, a_3 \ldots a_n (n \in N)$ is a sequence such that $a_1=a_2=2$ and $\frac{2a_{n-1}a_n}{a_{n-1}a_{n+1}-a_n^2}=n^3-n, \forall \mathrm{n}=2,3,4, \ldots$, find the value of $[\mathrm{S}]$, where $\mathrm{S}=\sum_{\mathrm{k}=2}^{2000} \frac{\mathrm{a}_{\mathrm{k}+1}}{\mathrm{a}_{\mathrm{k}}}$ and ([-] denotes G.I.F.)

Answers & Solutions

Q.1 Answer:

$3.6 \times 10^{-15}\ \mathrm{m}$

Solution:

The relation between radius (R) and atomic number (A) is

$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3}$

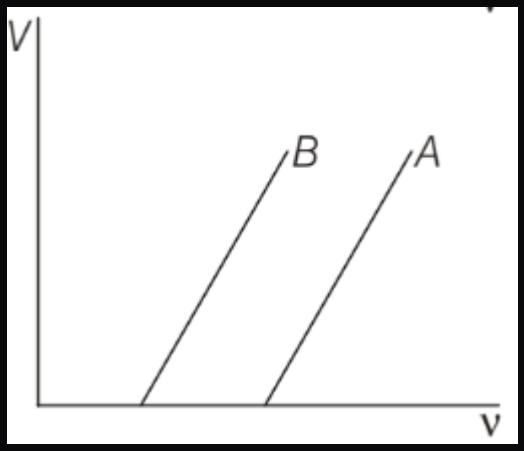
Given, $R_1 = 6$ fermi, $A_1 = 125, A_2 = 27$

$$\frac{6}{R_2} = \left(\frac{125}{27}\right)^{1/3} = \frac{5}{3}$$

$$\Rightarrow R_2 = \frac{6 \times 3}{5} = \frac{18}{5} = 3.6 \text{ fermi}$$

$$= 3.6 \times 10^{-15} \text{ m}$$

Q.2 Answer:



Solution:

Work function is proportional to y-intercept on V versus v graph. In case (d), line of A cuts larger intercept.

Q.3 Answer:

2 A

Solution:

Diode, D_1 = reverse biased (OFF)

and diode, D_2 = forward biased (ON).

Because diodes are ideal, so voltage drop across D_2 is zero.

$\therefore$ Effective resistance, $R = 3 + 2 + 3 + 2 = 10\Omega$

Current through the cell,

$I = V/R = 20/10 = 2$ A

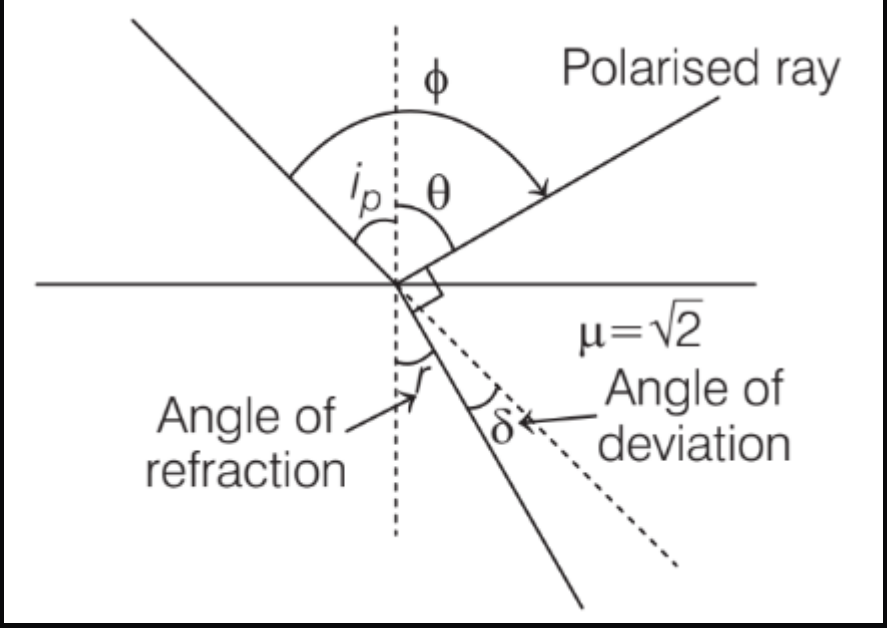
Q.4 Answer:

(iv) (iii) (i) (ii)

Solution:

Figure showing the ray diagram of refraction at a plane surface.

Given, i_p = angle of polarisation



So,

$$i_p = \tan^{-1}\mu \Rightarrow i_p = \tan^{-1}\sqrt{2} = \sin^{-1}\sqrt{\tfrac{2}{3}}$$

From Snell's law,

$$\sin i_p = \mu \sin r$$

Here, $\mu = 1.414 = \sqrt{2} \Rightarrow \sin r = \frac{\sin i_p}{\mu}$

$$\sin r = \frac{1}{\sqrt{2}}\frac{\sqrt{2}}{\sqrt{3}}$$

$$r = \sin^{-1}\frac{1}{\sqrt{3}}$$

angle of refraction

Now, the angle of deviation,

$$\delta = i_p - r$$

$$\delta = \sin^{-1}\sqrt{\frac{2}{3}} - \sin^{-1}\frac{1}{\sqrt{3}}$$

Angle of reflection,

$$\theta = 90^\circ - r \Rightarrow r = 90^\circ - \theta$$

$$\sin^{-1}\frac{1}{\sqrt{3}} = 90^\circ - \theta \Rightarrow \frac{1}{\sqrt{3}} = \sin(90^\circ - \theta)$$

$$\cos\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = \cos^{-1}\frac{1}{\sqrt{3}}$$

Angle between incident and completely polarised light is given by

$$\phi = i_p + \theta$$

$$\phi = \sin^{-1}\sqrt{\frac{2}{3}} + \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) = 2\sin^{-1}\sqrt{\frac{2}{3}}$$

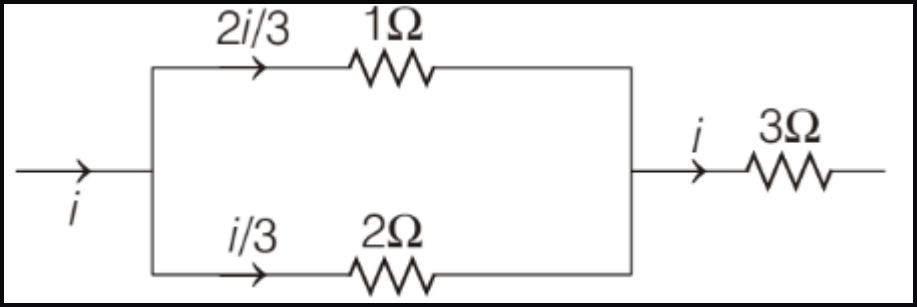
Hence, the correct option is (d).

Q.5 Answer:

4 : 2 : 27

Solution:

Current distribution is as shown in figure



∴ Powers are

$$P_1 = I_1^2 R_1 = \frac{4}{9} i^2, P_2 = I_2^2 R_2 = \frac{2}{9} i^2, \text{ and } P_3 = I_3^2 R_3 = 3i^2$$

$$= \frac{27}{9} i^2.$$

$$\Rightarrow \quad P_1 : P_2 : P_3 :: 4 : 2 : 27$$

Q.6 Answer:

30°

Solution:

Minimum force required to move a body up a rough inclined plane

$$F_1 = mg(\sin\theta + \mu\cos\theta)$$

Minimum force required to prevent the body from sliding down the rough inclined plane

$$F_2 = \mu mg\cos\theta$$

According to question,

$$F_1 = 3F_2$$

$$\therefore mg(\sin\theta + \mu\cos\theta) = 3(\mu mg\cos\theta)$$

$$\sin\theta + \mu\cos\theta = 3\mu\cos\theta$$

$$\sin\theta = 2\mu\cos\theta$$

$$\tan\theta = 2\mu$$

$$= 2 \times \frac{1}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$= \tan 30^\circ$$

$$\therefore \quad \theta = 30^\circ$$

Q.7 Answer:

$2v_e$

Solution:

Since velocity of projection (v) is greater than the escape velocity (v_e), therefore at infinite distance the body moves with a velocity

$$v' = \sqrt{v^2 - v_e^2}$$

$$\therefore \quad v' = \sqrt{\left(\sqrt{5}v_e\right)^2 - v_e^2} = 2v_e$$

Q.8 Answer:

3

Solution:

Given, speed of the electron in 1st Bohr's orbit,

$v_1 = 2.18 \times 10^6 \text{ m/s}$

Time period of nth orbit,

$T_n = 4.10\text{f} - \text{s} = 4.1 \times 10^{-15} \text{ s}$

Radius of Bohr's first orbit, $r_1 = 0.53 \times 10^{-10} \text{ m}$

$\therefore$ Orbital period of electron in Bohr's first orbit,

$$T_1 = \frac{2\pi r_1}{v_1} = \frac{2 \times 314 \times 0.53 \times 10^{-10}}{2.18 \times 10^6}$$

$$T_1 = 1.52 \times 10^{-16} \text{ s}$$

Again, Time period of nth orbit is given by

$$T_n = n^3 T_1$$

$$n^3 = \frac{T_n}{T_1} = \frac{4.1 \times 10^{-15}}{1.52 \times 10^{-16}} = 26.98$$

$$n^3 = 27 \Rightarrow n = (27)^{1/3} \Rightarrow n = 3$$

Q.9 Answer:

Peak voltage for AC is much larger

Solution:

Rms value of AC, $V_{\text{rms}} = 220 \text{ V}$

Peak value of AC, $V_0 = \sqrt{2} V_{\text{rms}} = \sqrt{2} \times 220 = 311 \text{ V}$

$V_{DC} = 220 \text{ V}$

Since, peak value of 220 V AC is 311 V which is much more than 220 DC. Hence, 220 V AC is more dangerous than 220 V DC.

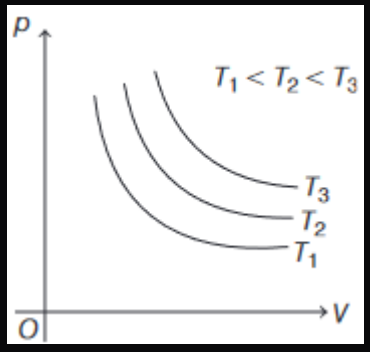
Q.10 Answer:

1, 4 are correct

Solution:

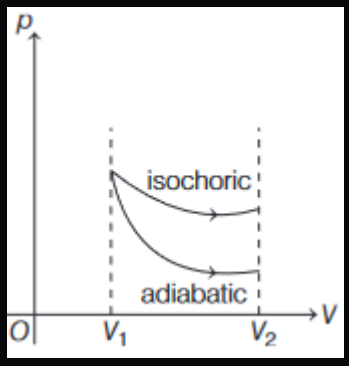
(i) and (iv) options are correct.

(i) Slope of $p - V$ graph in an isothermal process is $-\frac{p}{V}$ this is correct.



For isothermal process, slope of $p - V'$ curve $\frac{dp}{dV} = -\frac{p}{V}$

(ii) Slope of $p - V$ group in an adiabatic process is $-\frac{p}{V}$, which is wrong.

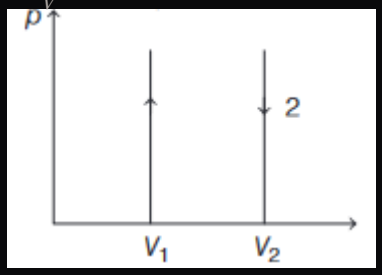


Because slope of $p - V$ graph in an adiabatic process is $-\frac{\gamma p}{V}$.

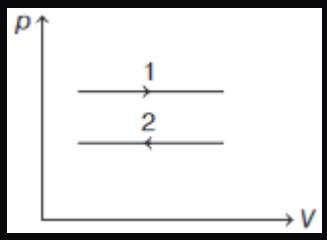
(iii) Slope of p - V graph in an isochoric process is $-\frac{\gamma p}{V}$ this option is wrong.

Because slope of $p - V$ graph for isochoric process is

$\frac{dp}{-\frac{dV}{V}} - \frac{dV}{V} = \frac{dp}{0} = \infty$



(d) Slope of $p - V$ graph in an isobaric process is zero. This option is correct.



$p - V$ curve is a straight line parallel to volume axis, slope of p - V curve for an isobaric process $\frac{dp}{dV} = 0$.

Q.11 Answer:

$\sin^{-1}\left(\frac{4}{5}\right)$

Solution:

$x = 36t$ metre

$2y = \left(96t - 9.8t^2\right)$ metre

Standard equations are,

$x = (u\cos\theta)t$

$y = u\sin\theta t - \frac{1}{2}gt^2$

$u\cos = 36$

$u\sin\theta = \frac{96}{2} = 48$

$\therefore \quad \tan\theta = \frac{u\sin\theta}{u\cos\theta} = \frac{48}{36} = \frac{4}{3}$

$\theta = \tan^{-1}\frac{4}{3}$

or $\theta = \sin^{-1}\frac{4}{5}$

Q.12 Answer:

stefan's constant

Solution:

Dimension of Boltzmann's constant

$A = \frac{\mathrm{J}}{\mathrm{K}} = \frac{\mathrm{ML^2\ T^{-2}}}{\mathrm{K^1}} = \left[\mathrm{ML^2\ T^{-2}\ K^{-1}}\right]$

Dimension of Planck constant

$B = \mathrm{Js} = \left[\mathrm{ML^2\ T^{-1}}\right]$

Dimension of speed of light,

$C = \left[\mathrm{LT^{-1}}\right]$

Desired dimension $= A^4B^{-3}C^{-2}$

$\Rightarrow \left[\mathrm{ML^2\ T^{-2}\ K^{-1}}\right]^4 \left[\mathrm{ML^2\ T^{-1}}\right]^{-3} \left[\mathrm{LT^{-1}}\right]^{-2}$

$\Rightarrow \left[\mathrm{M^1\ L^0\ T^{-3}\ K^{-4}}\right]$

Stefan constant (σ) **is given by**

$E = \sigma T^4 \Rightarrow \sigma = \frac{E}{T^4}$

where, $E =$ **energy per unit area per unit time and** $T =$ **absolute temperature.**

$\Rightarrow$ **Dimension,** $\sigma = \frac{\left[\mathrm{ML^2\ T^{-2}}\right]}{[\mathrm{L^2}][\mathrm{T}][\mathrm{K^4}]}$

$\sigma = \left[\mathrm{ML^0\ T^{-3}\ K^{-4}}\right]$

So, desired dimension is of Stefan constant.

Q.13 Answer:

21

Solution:

As $C_V = 12.6\ \mathrm{J\ mol^{-1}\ K^{-1}}$ **is given in SI unit**

So, $R = 8314\ \mathrm{J\ mol^{-1}\ K^{-1}}$

Then, $C_p = C_V + R = 12.6 + 8.314 = 21$

Q.14 Answer:

$\sqrt{5}p$

Solution:

If $p'\hat{\mathbf{n}}$ **be the momentum of third parts, then according to conservation of linear momentum. Total moment before explosion** $=$ **Total momentum after explosion**

$\Rightarrow 0 = -2p\hat{\mathbf{i}} + p\hat{\mathbf{j}} + p'\hat{\mathbf{n}}$

$\Rightarrow p'\hat{\mathbf{n}} = 2p\hat{\mathbf{i}} - p\hat{\mathbf{j}}$

$\therefore$ Magnitude of $p'\hat{\mathbf{n}} = p'$

$= \sqrt{(2p)^2 + (-p)^2} = \sqrt{5p^2} = \sqrt{5p}$

Q.15 Answer:

$Mg\sqrt{3}$

Solution:

By Newton's law

$$F = T\sin\theta$$

$$Mg = T\cos\theta$$

Dividing Eq. (i) by Eq. (ii), we get

$$\frac{F}{Mg} = \frac{T\sin\theta}{T\cos\theta}$$

$$F = Mg\tan\theta \quad (\because \theta = 60^\circ \text{ given})$$

$$F = Mg\tan 60^\circ$$

$$F = \sqrt{3}Mg$$

Q.16 Answer:

200

Solution:

Least count of screw gauge $= 5\,\mu\text{m}$

$$\text{L. C} = \frac{\text{pitch}}{\text{no. of div on circular scale}}$$

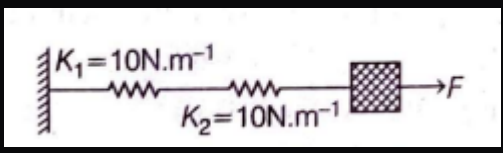
$$5\,\mu\text{m} = \frac{1\text{ mm}}{\text{N}}$$

$$\text{N} = 200$$

Q.17 Answer:

2500

Solution:



For series combination, effective force constant is given by

$$\frac{1}{K_{\text{eff}}} = \frac{1}{K_1} + \frac{1}{K_2}$$

Here, $K_1 = K_2 = 10\text{ N/m}$

$$\Rightarrow \quad \frac{1}{K_{\text{eff}}} = \frac{1}{10} + \frac{1}{10} = \frac{1}{5}$$

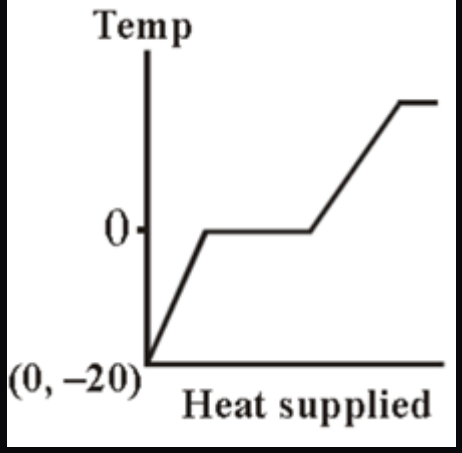
or $K_{\text{eff}} = 5\text{ N/m} = \frac{5\times10^5}{100}\text{dyne/cm}$

$$\Rightarrow \quad K_{\text{eff}} = 5\times10^3 \text{ dyne/cm}$$

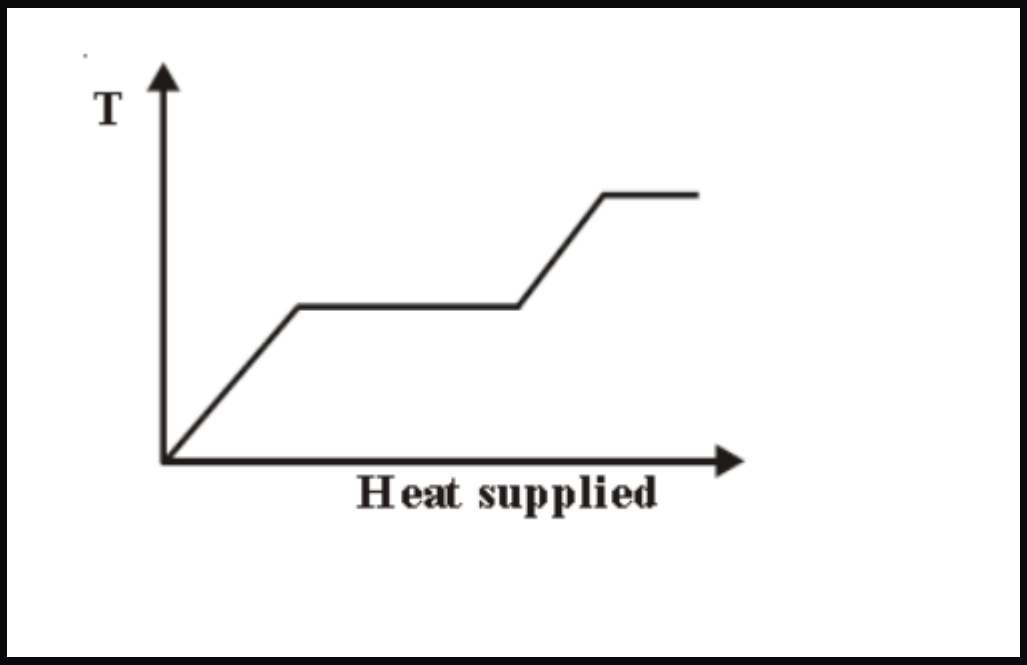
Work done in stretching this spring,

$$W = \frac{1}{2}K_{\text{eff}}x^2 = \frac{1}{2}\times5\times10^3\times1^2 = 2500\text{erg}$$

Q.18 Answer:



Solution:



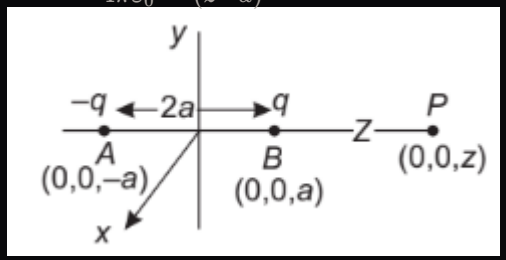
Q.19 Answer:

$\frac{2qa}{4\pi\varepsilon_0(z^2-a^2)}$

Solution:

Potential at P due to $(+q)$ charge

$V_1 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{(z-a)}$



Potential at P due to $(-q)$ charge

$V_2 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{-q}{(z+a)}$

Total potential at P due to (AB) electric dipole

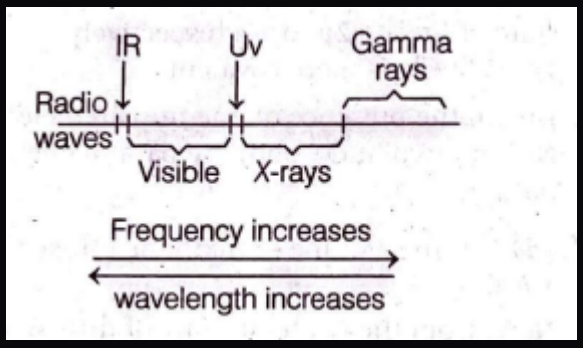
$$V = V_1 + V_2$$
$$= \frac{1}{4\pi\varepsilon_0}\cdot\frac{q}{(z-a)} - \frac{1}{4\pi\varepsilon_0}\frac{q}{(z+a)}$$
$$= \frac{q}{4\pi\varepsilon_0}\frac{(z+a-z+a)}{(z-a)(z+a)}$$
$$\Rightarrow \quad V = \frac{2qa}{4\pi\varepsilon_0(z^2-a^2)}$$

Q.20 Answer:

Gamma ray, X-ray, ultraviolet, visible light, infrared.

Solution:

EM spectrum is as shown



So, correct order of ascending wavelength is Gamma rays $\rightarrow$ X-rays $\rightarrow$ UV Infrared $\rightarrow$ Visible light $\rightarrow$ Infrared

Q.21 Answer:

—

Solution:

Hint:

$I_1 = 1.51_2$

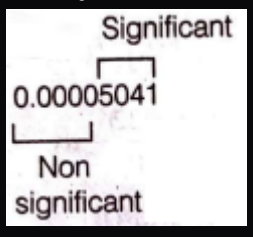
$\frac{I_1}{I_2} = \frac{3}{2}\cdot \quad \frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{L_1}+\sqrt{L_2}}{\sqrt{I_1}-\sqrt{L_2}}\right)^2 = \left(\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}\right)^2 = 98$

Q.22 Answer:

—

Solution:

In any number less than 1, leading zeros before or after decimal are non significant. So in given case,



Number of significant digits is 4.

Note **Any zero between two non zero digits is always significant.**

Q.23 Answer:

—

Solution:

$A = 60°, \delta = 30°$

Refractive index of prism with respect to liquid,

$$\mu = \frac{\sin\frac{A+\delta_n}{2}}{\sin\frac{A}{2}} = \frac{\sin\frac{(60°+30°)}{2}}{\sin\frac{60°}{2}}$$

$$= \frac{\sin 45°}{\sin 30°} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} = \sqrt{2}$$

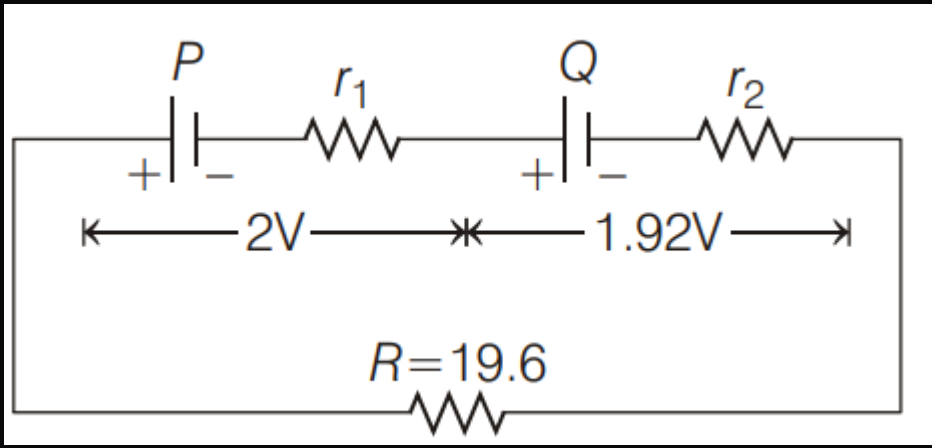
$\therefore$ **Critical angle** $= \sin^{-1}\left(\frac{1}{\mu}\right) = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$

$= \sin^{-1}(\sin 45°) = 45°$

Q.24 Answer:

—

Solution:



Emf of each cell $= 2.16$ V

Emf of circuit $= 4.32$ V

Current in circuit,

$I = \frac{4.32}{r_1+r_2+19.6}$

Now, using $V = E - I \cdot r$ **for both cells, we have**

For P, $2 = 2.16 - \frac{4.32r_1}{r_1+r_2+19.6}$

and For Q, $1.92 = 216 - \frac{4.32r_2}{r_1+r_2+19.6}$

So, we have,

$$0.16 = \frac{4.32r_1}{r_1+r_2+19.6} \quad \ldots(i)$$

$$\text{and,} \quad 0.24 = \frac{4.32r_2}{r_1+r_2+19.6} \quad \ldots(ii)$$

From Eqs. (i) and (ii), we get

$r_1 : r_2 = 2 : 3$

Q.25 Answer:

—

Solution:

Using $\mathrm{E} = \frac{dV}{dr}$ or, $\mathrm{dV} = \mathrm{Edr}$

Displacement, $\mathrm{dr} = \mathrm{r_B} - \mathrm{r_A}$

$= (2\hat{i} + \hat{j} + 3\hat{k}) - (\hat{i} + 2\hat{j})$

$= (10\hat{i} + 30\hat{j})(\hat{i} - \hat{j} + 3\hat{k})$

$= 10\hat{i} \cdot \hat{i} - 30\hat{i} \cdot \hat{j} + 0$

$= 10 - 30 = -20$

Work done, $\mathrm{W} = \mathrm{q} \times |\mathrm{dV}| = 0.8(20) = 16\ \mathrm{J}$

Q.26 Answer:

only I is correct

Solution:

The physical and chemical properties of elements are periodic functions of their electronic configuration. This is the correct statement.

Q.27 Answer:

$\mathrm{CuCl_2}$

Solution:

Copper chloride $(\mathrm{CuCl_2})$ as a yellow-brown powder (in anhydrous form) or a green crystalline solid (in dihydrate form) due to copper exist as divalent $\mathrm{Cu^{2+}}$ ions with $3d^9$ electronic configuration. Due to presence of 1 unpaired electron in $3d$-orbital, it is coloured.

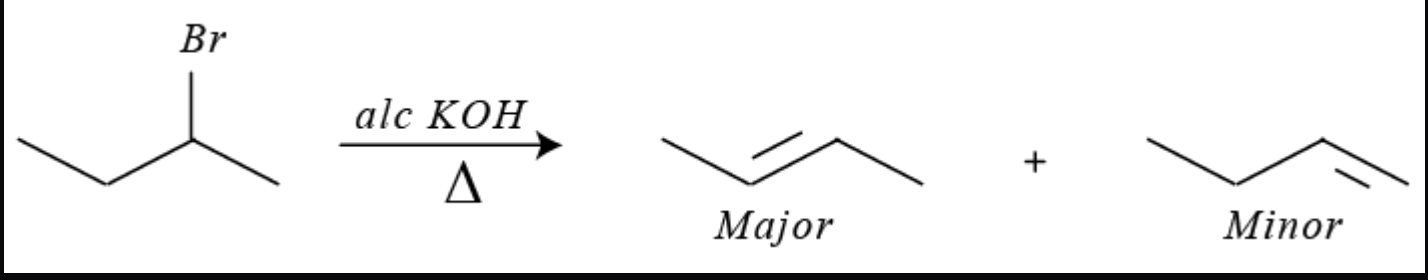
Remaining $\mathrm{ZnSO_4}$, $\mathrm{MgCl_2}$ and AgCl are not coloured compounds respectively.

Hence, only $\mathrm{CuCl_2}$ is expected to be coloured.

Q.28 Answer:

b,d

Solution:

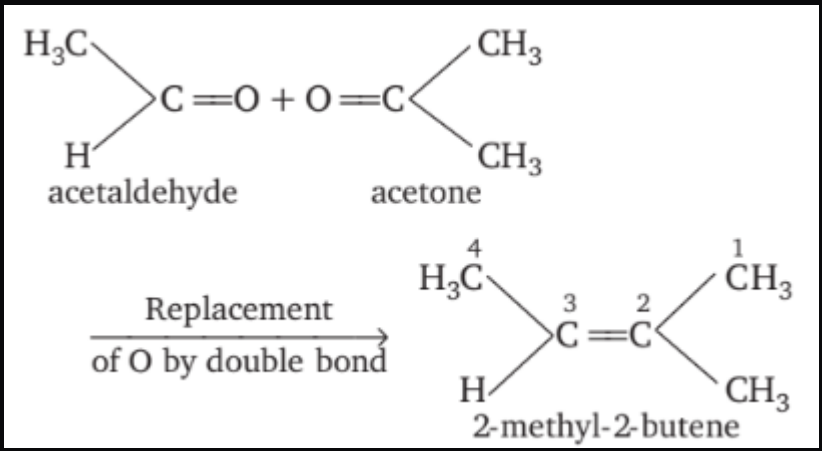


Q.29 Answer:

2-methyl-2-butene

Solution:

To decide the structure of alkene that undergoes ozonolysis, bring the products together in such a way that O atoms are face to face, and replace O by double $(=)$ bond. Thus,



Q.30 Answer:

0.3 K

Solution:

$\Delta T_f = K_f \times \frac{w_1 \times 1000}{w_2 \times m_1}$

$w_1 =$ weight of solute

$w_2 =$ weight of solvent

$m_1 =$ molar mass of solute

$K_f = 4.3 \text{ K kg mol}^{-1}$

Now, $\Delta T_f = 4.3 \times \frac{2 \times 1000}{57.3 \times 500}$

$= \frac{17.2}{57.3} = 0.30 \text{ K}$

Q.31 Answer:

$3s^2 3p^3$

Solution:

Elements having half or completely-filled orbitals, ie, having stable electronic configuration have very low negative value of electron affinity.

Since, the electron affinity of B is lowest among the given, it has stable electronic configuration,

$ie, 3s^2, 3p^3.$

Q.32 Answer:

18

Solution:

According to question,

$w_A = x \text{ g}, m_A = 18, X_A = 1 - 0.6 = 0.4$

$w_B = 69 \text{ g}, m_B = 46, X_B = 0.4$

We know that,

$X_A = \frac{n_A}{n_A + n_B}$

$$\text{or} \quad 0.4 = \frac{w_A/m_A}{w_A/m_A + 69/46} \Rightarrow 0.4 = \frac{\frac{x}{18}}{\frac{x}{18} + \frac{3}{2}}$$

$$0.4 \times \left(\frac{2x + 54}{36}\right) = \frac{x}{18}$$

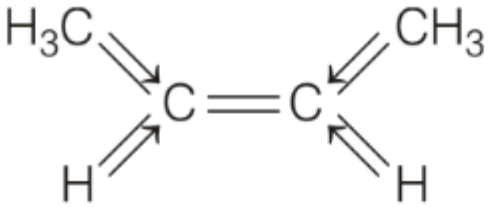
or $2x + 54 = 5x$ **or** $3x = 54, x = 18g$

Q.33 Answer:

cis-1, 2-dichloroethene

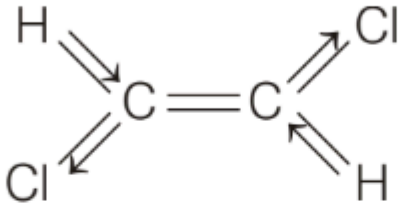
Solution:

(c) (a) *cis*-but-2-ene



(μ=0.33 D)

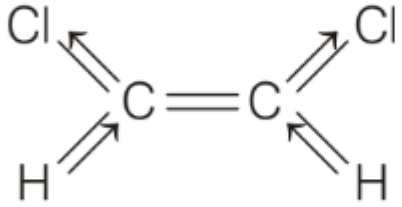
(b) *trans*-1, 2-dichloroethene



(μ=0 D)

(Bond moments cancel out in *trans* form)

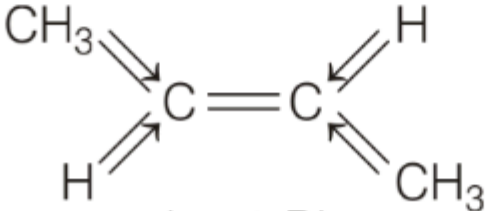
(c) *cis*-1, 2-dichloroethene



(μ=1.89 D)

Bond moments do not cancel out in *cis* form)

(d) *trans*-but-2-ene



(μ=0 D)

Q.34 Answer:

$C_6H_{10}O_2$ N

Solution:

The general formula for this series is C_nH_{2n-2}. Hence, $C_6H_{10}O_2$ N belongs to the given homologous series.

Q.35 Answer:

Mn^{2+} only

Solution:

$$2K\overset{+7}{Mn}O_4 + 5SO_2 + 2H_2O \longrightarrow K_2SO_4 + 2 + \overset{+2}{Mn}SO_4 + 2H_2SO_4$$

Q.36 Answer:

$[NiCl_4]^{2-}$

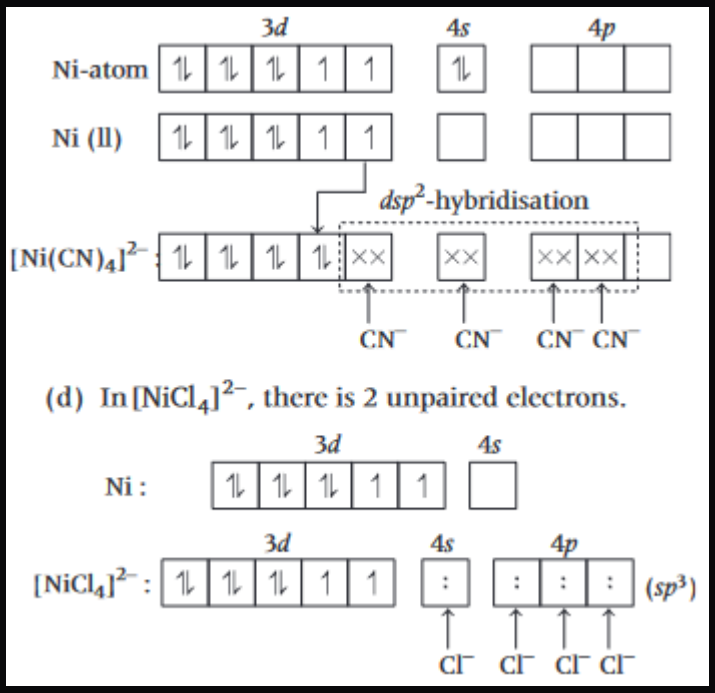
Solution:

(a) $[Fe(CN)_6]^{3-}$ has d^2sp^3 (inner d - complex) hybridisation with one electron unpaired.

(b) The Fe^{2+} ion in $[Fe(CN)_6]^{4-}$ has outer electronic configuration of $3d^6$. The complex is a low spin complex. It contains 0 unpaired electrons with magnetic moment of $0BM$ Therefore, under the influence of the octahedral crystal field, the possible electronic arrangement of Fe (II) ion is t_{2g}^6, e_g^0.

(c) $[Ni(CN)_4]^{2-}$ is a square planar geometry formed by dsp^2 hybridisation. $[Ni(CN)_4]^{2-}$ is diamagnetic, so Ni^{2+} ion has $3d^8$, outer configuration with two unpaired electrons.

For the formation of the square planar sructure by structure by dsp^2 - hybridisation, two unpaired d-electrons are paired up due to energy made available by the approach of ligands, making one of the $3d$-orbitals empty.



$\therefore Cl^-$ is a weak ligand so, there is no pairing of electrons.

Number of unpaired $= 2$

Hence, the correct option is (d).

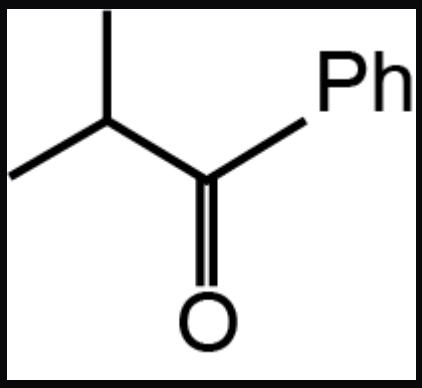
Q.37 Answer:

(A) is true, (R) is true and (R) is the correct explanation for (A)

Solution:

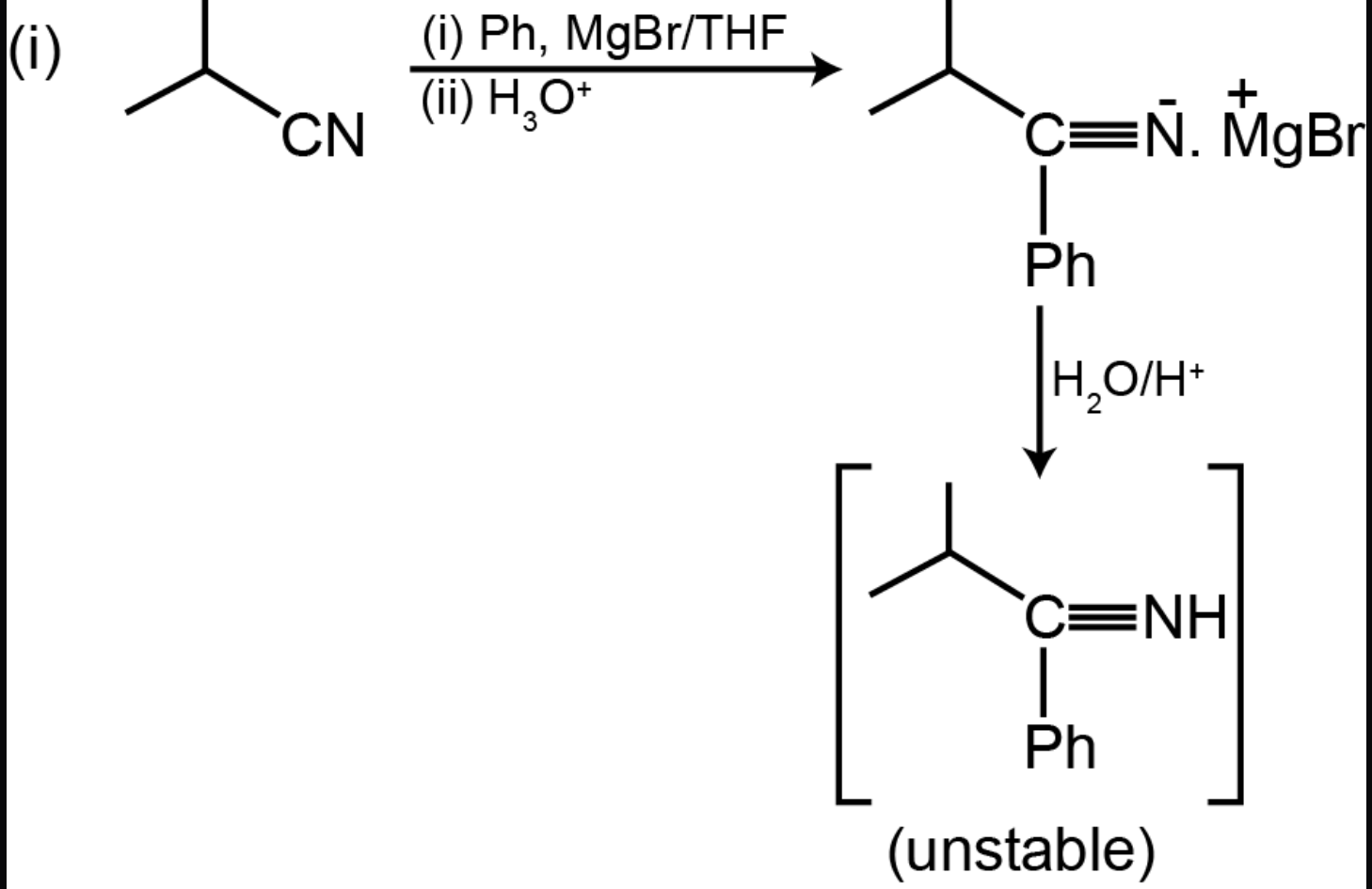
We know that fluorine atom can stabilise the higher oxidation state due to its higher melting point, higher bond enthalpy and small size and that is the reason with vanadium it forms VF_5 whereas VCl_5 is unstable due to the large size of chlorine atom such that there is no place of accommodation for five big chlorine atoms near the central metal vanadium and because of this stearic hindrance VCl_5 does not exist in isolated form while it can exist in dimer form in solid phase.
Hence, assertion and reason both are true and reason is correct explanation of assertion

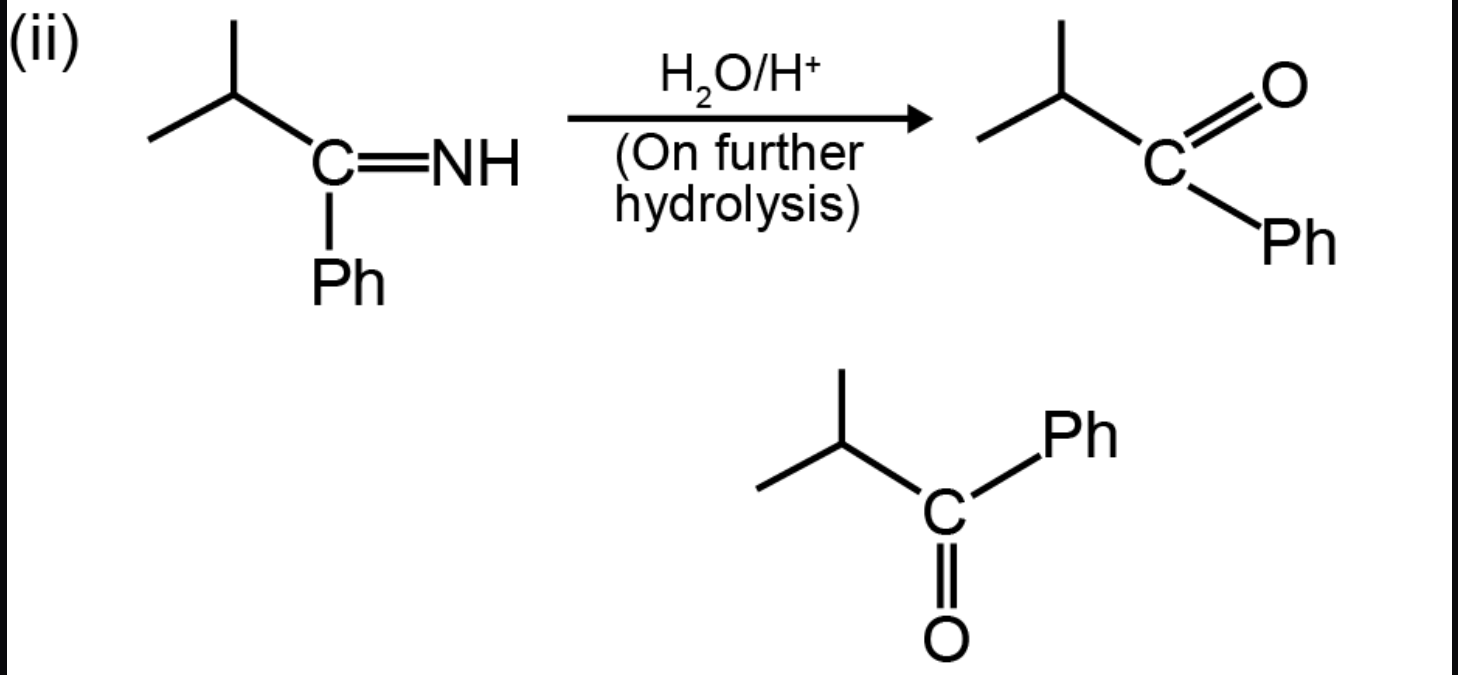
Q.38 Answer:



Solution:

The given reaction occurs as follows :





Q.39 Answer:

$Mg_2P_2O_7$

Solution:

In organic compound, phosphorus can be estimate by precipitating it as $MgNH_4PO_4$, by adding magnesia mixture. On ignition of precipitate, it yields $Mg_2P_2O_7$.

Q.40 Answer:

The cell reaction is $Zn + Cu^{+2} \longrightarrow Zn^{+2} + Cu$

Solution:

In the Daniell cell, copper and zinc electrodes are immersed in a solution of copper(II) sulphate and zinc sulphate, respectively.

At the anode (negative electrode), zinc is oxidised per the following half reaction.

$$\mathrm{Zn}(s) \longrightarrow \mathrm{Zn}^{2+}(aq) + 2e^{-}$$

At the cathode (positive electrode), copper is reduced per the following reaction.

$$\mathrm{Cu}^{2+}(aq) + 2e^{-} \longrightarrow \mathrm{Cu}(s)$$

Overall cell reaction is

$$\mathrm{Zn}(s) + \mathrm{Cu}^{2+}(aq) \longrightarrow \mathrm{Zn}^{2+}(aq) + \mathrm{Cu}(s)$$

These processes result in the accumulation of solid copper at the cathode and the corrosion of the zinc electrode into the solution as zinc cations. Hence, the correct option is (d).

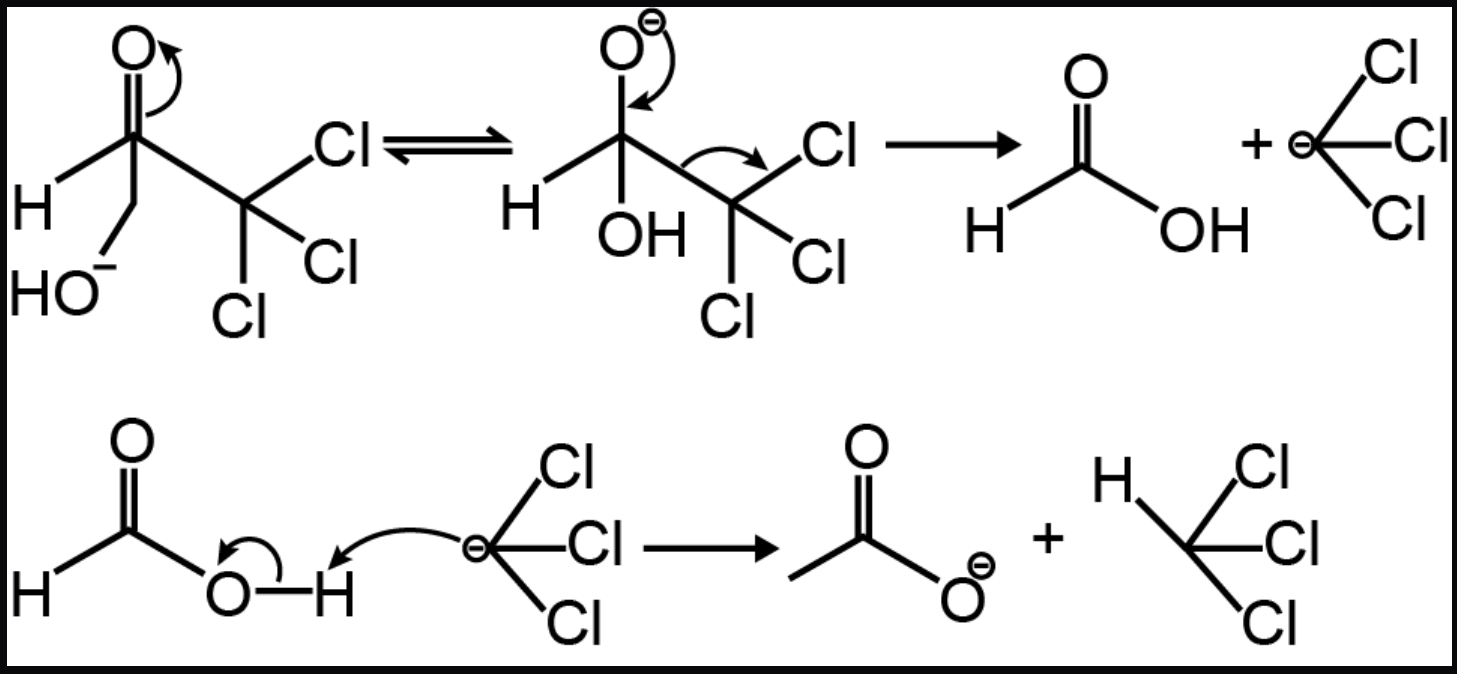
Q.41 Answer:

$\mathrm{Cl_3CCHO}$

Solution:

For a carbonyl group to give Cannizaro reaction, it should not have an α -hydrogen atom. Among the given compound $\mathrm{Ne_3CCHO}$, $\mathrm{C_6H_5CHO}$ and HCHO respond

to Cannizaro reaction but $\mathrm{Cl_3CCHO}$ do not due to the following reason.



When hydroxyl group attacks the carbonyl group a tetrahedral intermediate forms. This tetrahedral intermediate will revert to a carbonyl compound by

expelling the best leaving group. The $\mathrm{CCl_3}$ carbonium is resonance stabilised therefore it is better leaving

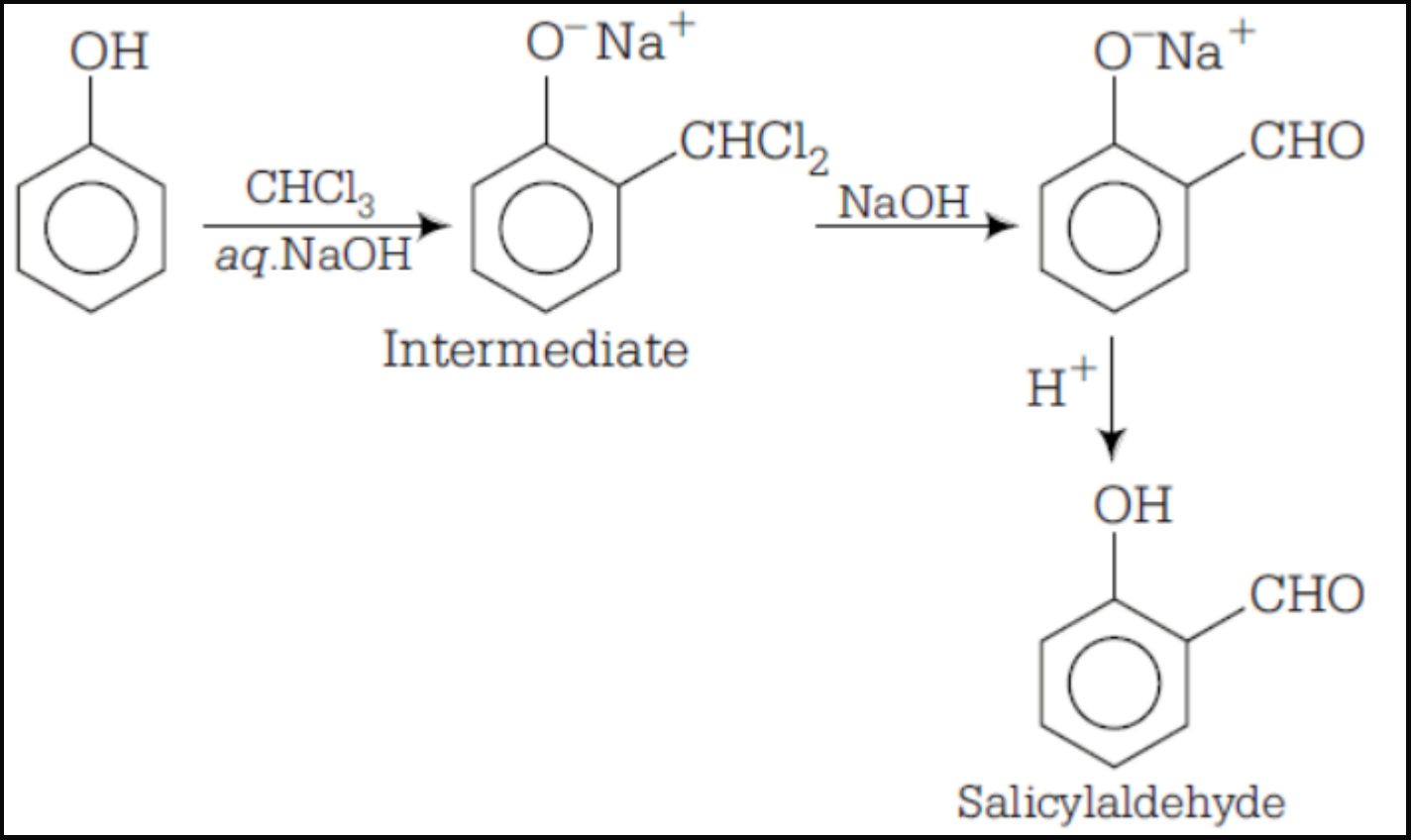
group than OH. These initial products exchange a proton to give ion of carboxylic acid and trichloromethane ($CHCl_3$).

Q.42 Answer:

Substituted benzal chloride

Solution:

In Reimer-Tiemann reaction, on treating phenol with chloroform in the presence of sodium hydroxide, a $-CHO$ group is introduced at ortho position of benzene ring.



Hence, substituted benzyl chloride is intermediate in Reimer-Tiemann reaction.

Q.43 Answer:

5 and 5.92

Solution:

Ferrous ion $\left(Fe^{2+}\right)$ changes to ferric ion Fe^{3+} on reacting with acidified H_2O_2 as

$2\ K_4\left[Fe(CN)_6\right] + H_2SO_4 + H_2O_2 \longrightarrow 2\ K_3\left[Fe(CN)_6\right] + K_2SO_4 + 2H_2O$

Electronic configuration of $Fe^{3+} =$

$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^5$

Number of d electrons $= 5$

Magnetic moment $= \sqrt{n(n+2)}$

$= \sqrt{5(5+2)} = 5.92BM$

Q.44 Answer:

$3 < 1 < 4 < 2$

Solution:

S_N1 reaction is a two step reaction in which greater the stability of carbocation, greater will be its formation from alkyl halide and faster will be the rate of reaction. As, the stability of carbocations formed is in the order.

$(2) > (4) > (1) > (3)$

$\therefore$ The order of reactivity towards S_N1 reaction will be $(3) < (1) < (4) < (2)$.

Q.45 Answer:

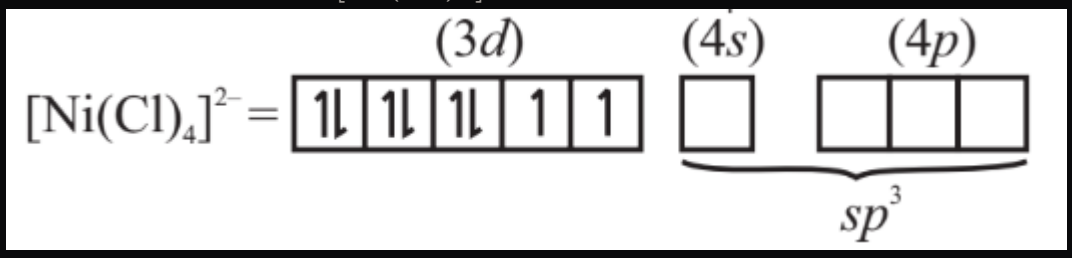
$[Ni(CO)_4]$

Solution:

(a) Ni in $[Ni(Cl)_4]^{2-}$ exist as Ni^{2+} ion.

$\because$ $C\ell$ is a weak field ligand (high spin). It will not cause pairing of electrons.

Hence, configuration of $[Ni(C\ell)_4]^{2-}$ is

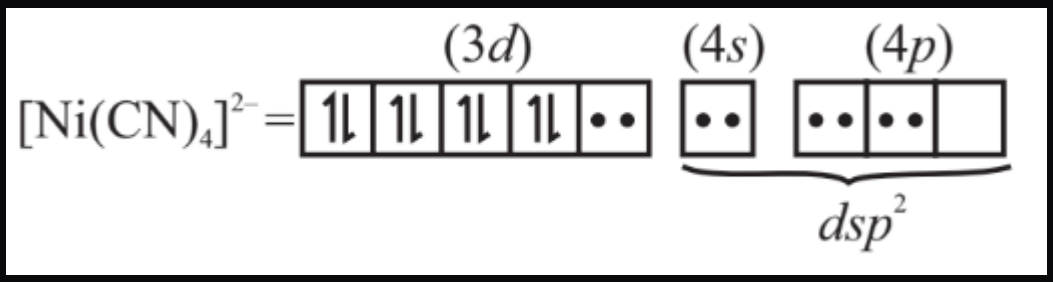


Tetrahedral with two unpaired electrons (i.e., paramagnetic) and has sp^3 hybridisation.

(b) In $[Co\,(C_2O_4)_3]$, $(C_2O_4)_3$ is a bidentate ligand thus, give octahedral structure.

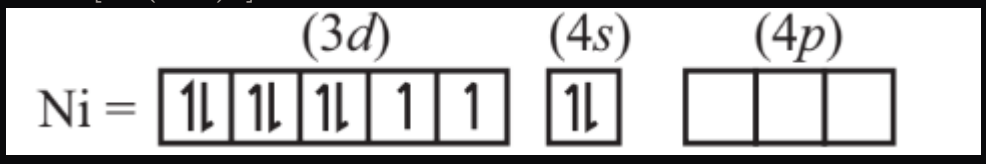
(c) In $[Ni(CN)_4]^{2-}$, Ni exist as Ni^{2+} ion.

$\because$ CN^- is a strong field ligand (low spin), causes pairing of electrons of $3d$ orbital.

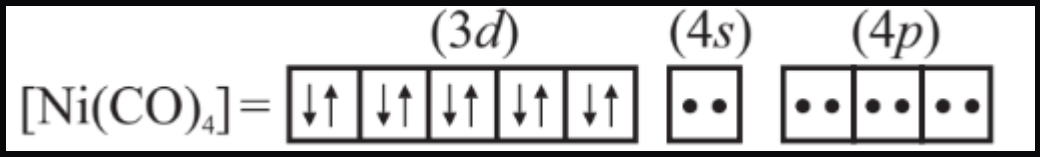


Hence, the structure of $[Ni(CN)_4]^{2-}$ is square planar and it is diamagnetic.

(d) In $[Ni(CO)_4]$, Ni has zero oxidation state. i.e.,



CO is a strong field ligand causes rearrangement and pairing of electrons of $3d$ and $4s$ orbital.



Structure of $Ni(CO)_4$ is tetrahedral with sp^3 hybridisation and it is diamagnetic.

Hence, (d) is the correct answer.

Q.46 Answer:

—

Solution:

$\because$ If CO_3^{2-}, SO_3^{2-} and SO_4^{2-} are present alongwith $BaCl_2$, these can also show white precipitate (as precipitate of all these are also white).

Q.47 Answer:

—

Solution:

$N_2H_4 - 10e^- \longrightarrow X^{10+}$ (with two N atoms)

Total oxidation number of two N atoms in N_2H_4,

$\Rightarrow \quad 2x + 4 = 0$

$2x = -4$

N_2H_4 loses 10 electrons, so total oxidation number of two N-atoms increases by 10 , i.e., the total oxidation number of two N-atoms in

$Y = -4 + 10 = +6$

$\therefore$ Oxidation number of each N atom in

$X^{10+} = +3$

Q.48 Answer:

—

Solution:

The total number of electron exchanges possibility is $\frac{n(n-1)}{2}$ (where, n is the number of shells).

By applying this formula for d^5 configuration we get,

$= \frac{5(5-1)}{2} = \frac{5\times4}{2} = \frac{20}{2} = 10$

Q.49 Answer:

—

Solution:

Hydrolysis of sucrose give glucose and fructose. Hydrolysis of sucrose brings about a change in the sign of rotation, from dextro (+) to laevo (–). Hence, Sucrose is called invert sugar.

$C_{12}H_{22}O_{11}(\text{Sucrose}) \ + \ H_2O \rightarrow C_6H_{12}O_6(\text{Glucose}) + C_6H_{12}O_6(\text{Fructose})$

In the above reaction one mole of sucrose is giving one mole of glucose. Hence the mass of sucrose is

$342\text{g}(\text{Molar mass})$

Q.50 Answer:

—

Solution:

For the reaction:

$$A \rightarrow \left\{ \begin{array}{l} 75\% \rightarrow 2\,\mathrm{B} + \mathrm{C} : \Delta \mathrm{H}_1 = 100\mathrm{cal/mol} \rightarrow 75\mathrm{cal} \\ 25\% \rightarrow -\mathrm{D} + \mathrm{E} : \Delta \mathrm{H}_2 = 200\mathrm{cal/mol} \rightarrow 50\mathrm{cal} \end{array} \right\} 125\mathrm{cal}$$

Let's take 100 particles of A initially

if 25 particles follow the lower $R_e n$ & make $D + E$ then

$[D] = 25$

75 partich -------- upper --------- $2B + C$

$[B] = 150$

Q.51 Answer:

19

Solution:

$$\frac{\left(1-x^2\right)^3\left(1-x^3\right)\left(1-x^4\right)}{(1-x)^5}$$

$$= \frac{(1-x)^3(1+x)^3(1-x)\left(1+x^2+x\right)(1-x)(1+x)\left(1+x^2\right)}{(1-x)^5}$$

$$= (1+x)^4\left(x^4+x^3+2x^2+x+1\right)$$

$$= {}^4C_1 + {}^4C_2 + 2^4C_3 + {}^4C_4 = 19$$

Q.52 Answer:

111

Solution:

$\log_6(abc) = 6 \rightarrow abc = 6^6$

$b^3 = 6^6 \Rightarrow b = 36$

$b^3 = 6^6 \Rightarrow b = 36$

For $\mathrm{a} = 27\ \&\ \ \mathrm{b} = 36 \Rightarrow \mathrm{c} = 48$ (only possibility)

so, $a + b + c = 111$

Q.53 Answer:

$\frac{1}{\sqrt{52}}$

Solution:

$(6 + r\cos\theta, 8 + r\sin\theta)$ lies on $2x^2 + y^2 = 2$

$\Rightarrow \left(2\cos^2\theta + \sin^2\theta\right)\mathrm{r}^2 + 2(12\cos\theta + 8\sin\theta)\mathrm{r} + 134 = 0$

$\mathrm{AB}, \mathrm{AP}, \mathrm{AC}$ are in $\mathrm{H} \cdot \mathrm{P} \Rightarrow \frac{2}{\mathrm{r}} = \frac{\mathrm{AB}+\mathrm{AC}}{\mathrm{AB}\cdot\mathrm{AC}} \Rightarrow \frac{1}{\mathrm{r}} = -\frac{(6\cos\theta + 4\sin\theta)}{67} \Rightarrow 6\mathrm{x} + 4\mathrm{y} - 1 = 0$

Minimum distance from $' \mathrm{O}' = \frac{1}{\sqrt{52}}$

Q.54 Answer:

$\frac{1}{5}$

Solution:

$$\left(\frac{3x+4y}{\sqrt{3^2+4^2}}\right)^2=\frac{1}{5}\left(\frac{4x-3y}{\sqrt{4^2+3^2}}\right)$$
$$LR=\frac{1}{5}$$

Q.55 Answer:

$\mathrm{A}-\mathrm{I}$

Solution:

$A^2-2A+21=0$
Divide B by $A^2-2A+21$
$\Rightarrow B=\left(A^2-2A+21\right)(f(A))+A-1$
$\Rightarrow B=A-I=\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}$

Q.56 Answer:

$xy-2x-4y=0$

Solution:

Let the equation of $AB\frac{x}{a}+\frac{y}{b}=1-(1)$. Let $p(h,k)$ locus of midpoint of $AB.a=2h,b=2k$ substitute in (1)
$\Rightarrow \frac{x}{2h}+\frac{y}{2k}=1\Rightarrow \frac{1}{2}(kh-hy)=hk$ and passes through the point $(x,y)=(8,4)$
$\Rightarrow \frac{1}{2}(8k+4h)=hk\Rightarrow 2h+4k=hk\Rightarrow 2x+4y=xy.$
The locus of p is $xy-2x-4y=0$

Q.57 Answer:

$-4\hat{\mathrm{i}}+3\hat{\mathrm{j}}-7\hat{\mathrm{k}}$

Solution:

Given $\vec{a}=2\hat{i}+\hat{j}-\hat{k},\vec{b}=\hat{i}+3\hat{j}-5\hat{k}$
Now, $3\overrightarrow{\mathrm{a}}-2\overrightarrow{\mathrm{b}}=3(2\hat{\mathrm{i}}+\hat{\mathrm{j}}-\hat{\mathrm{k}})-2(\hat{\mathrm{i}}+3\hat{\mathrm{j}}-5\hat{\mathrm{k}})$
$=4\hat{\mathrm{i}}-3\hat{\mathrm{j}}+7\hat{\mathrm{k}}$
Since $\overrightarrow{\mathrm{r}}$ along the vector $3\overrightarrow{\mathrm{a}}-2\overrightarrow{\mathrm{b}}$
So, $\vec{r}=x(3\vec{a}-2\vec{b})=x(4\hat{i}-3\hat{j}+7\hat{k})$
Now, $|\vec{r}|=\sqrt{74}\Rightarrow\sqrt{16x^2+9x^2+49x^2}=\sqrt{74}$
$\Rightarrow\sqrt{74}x^2=\sqrt{74}\Rightarrow x=\pm1$
since, $\vec{r}$ is the opposite direction of $3\vec{a}-2\vec{b}$
So $\mathrm{x}=-1$
$\Rightarrow\overrightarrow{\mathrm{r}}=-4\hat{\mathrm{i}}+3\hat{\mathrm{j}}-7\hat{\mathrm{k}}$

Q.58 Answer:

$f(1)+g(2)=6$

Solution:

(A)

$$f\left(\frac{1}{3}\right)=\left(\frac{1}{3}\right)^{1/3};g\left(-\frac{2}{3}\right)=\left(-\frac{2}{3}\right)^{2/3}$$

$$f\left(\frac{1}{3}\right)\cdot g\left(-\frac{2}{3}\right)=\frac{(-2)^{2/3}}{3^1}=\frac{3\sqrt{4}}{3}$$

(B)

$\mathrm{x}=\frac{\mathrm{p}+1}{\mathrm{q}}=\frac{\text{even}}{\text{odd}}$

$g(x)=x^{\left|\frac{\text{even}}{\text{odd}}\right|}=(x^{\text{even}})^{\frac{1}{\text{odd}}}$

$\therefore g(-x)=g(x)$

$\Rightarrow \mathrm{g(x)}$ **is even function**

(C)

$f(1)=1;g(2)=2^{|2|}=4$

$\therefore f(1)+g(2)=5$

(D)

$$\left(g\left(-\frac{6}{5}\right)\right)^{5/2}=\left(\left(-\frac{6}{5}\right)^{6/5}\right)^{5/2}=\left(\frac{6}{5}\right)^3$$

$$\left(g\left(\frac{4}{5}\right)\right)^{15/4}=\left(\left(\frac{4}{5}\right)^{4/5}\right)^{13/4}=\left(\frac{4}{5}\right)^3$$

$$\left(f\left(\frac{3}{5}\right)\right)^{5}=\left(\left(\frac{3}{5}\right)^{3/5}\right)^{5}=\left(\frac{3}{5}\right)^3$$

$\therefore \left[g\left(\frac{-6}{5}\right)\right]^{5/2}-\left[g\left(\frac{4}{5}\right)\right]^{15/4}-\left[f\left(\frac{3}{5}\right)\right]^5=1$

Q.59 Answer:

3

Solution:

$\overrightarrow{OG_1}\cdot\overrightarrow{BG_2}=0$

$\Rightarrow \frac{\vec{a}+\vec{b}+\vec{c}}{3}\cdot\frac{a+\vec{c}-3\vec{b}}{3}=0$

$\Rightarrow a^2+c^2-3b^2+2\vec{a}\cdot\vec{c}-2\vec{b}\cdot\vec{c}-2\vec{a}\cdot\vec{b}=0$

Now, $|\vec{c}-\vec{a}|^2=b^2,|\vec{c}-\vec{b}|=a^2$ **and** $|\vec{a}-\vec{b}-|=c^2$

$2\vec{a}\cdot\vec{c}=a^2+c^2-b^2,2\vec{b}\cdot\vec{c}$

$=b^2+c^2-a^2,2\vec{a}\cdot\vec{b}a^2b^2-c^2$

Putting in the above result, we get

$2a^2+2c^2-6b^2=0\quad\Rightarrow\frac{a^2+c^2}{b^2}=3$

Q.60 Answer:

only one minimum

Solution:

$$P(x) > 0$$

$$P'(x) = 2a_1x + 4a_2x^3 + \ldots = 2x\left(a_1 + 2a_2x^2 + \ldots\right)$$

$P'(x) = 0 \Rightarrow x = 0$, **since the second factor is positive.**

$\therefore P(x)$ **has minimum at** $x = 0$ **and** $P_{\min} = a_0$.

Q.61 Answer:

not defined

Solution:

$$f(x) = \begin{cases} \left(p^2 - 1\right)(x - 2) - 2; & -2 < x < -1 \\ -2\left(p^2 - 1\right) - 2; & x = -1 \\ q\left(\frac{e^x + e^{-x}}{2}\right) + |p|(x - 1); & -1 < x < 2 \end{cases}$$

$$-3\left(p^2 - 1\right) - 2 = -2\left(p^2 - 1\right) - 2$$

$$= \mathrm{q}\left(\frac{\mathrm{e}^{-1} + \mathrm{e}^{\mathrm{l}}}{2}\right) - 2|\mathrm{p}| \Rightarrow \mathrm{p} = \pm 1, \mathrm{q} = 0$$

$$f(x) = \begin{cases} -2; & -2 < x < -1 \\ -2; & x = -1 \\ x - 1; & -1 < x < 2 \end{cases}$$

$$f\left(f\left(f\left(\frac{-1}{2}\right)\right)\right) = f\left(f\left(\frac{-3}{2}\right)\right) = f(-2)$$

$$= \text{ Not defined}$$

Q.62 Answer:

24

Solution:

$$y = 10\sin^2 x + 6\sin x\cos x + 14\cos^2 x$$

$$= 10 + 4\cos^2 x + 3\sin 2x$$

$$= 10 + 2(1 + \cos 2x) + 3\sin 2x$$

$$= 3\sin 2x + 2\cos 2x + 12$$

$$y_{\min} = -\sqrt{3^2 + 2^2} + 12 = -\sqrt{13} + 12$$

$$y_{\max} = \sqrt{13} + 12$$

$$y_{\min} + y_{\max} = 24$$

Q.63 Answer:

5

Solution:

Let A : First 3 cards are king

B : Fourth card is a king

$\mathrm{P(A/B)} = \frac{\mathrm{P(A \cap B)}}{\mathrm{P(B)}}$

Let k : drawing of a king

B can occur in four types

Type 1 : $\overline{\mathrm{kkkk}} \to 1$

Type 2 : $k\bar{k}\bar{k}k \to 3$

Type 3 : $\mathrm{k}\bar{k}k\mathrm{k} \to 3$

Type 4 : $\mathrm{kkkk} \to 1$

$$P(A/B) = \frac{\frac{1}{{}^{52}\mathrm{C}_4}}{\frac{{}^{48}\mathrm{C}_3 \times {}^4\mathrm{C}_1}{4 \times {}^{52}\mathrm{C}_4} + \frac{{}^{48}\mathrm{C}_2 \times {}^4\mathrm{C}_2 \times 3}{{}^{52}\mathrm{C}_4 \times 6} + \frac{{}^{48}\mathrm{C}_1 \times {}^4\mathrm{C}_3 \times 3}{{}^{52}\mathrm{C}_4 \times 4} + \frac{1}{{}^{52}\mathrm{C}_4}}$$

$$= \frac{1}{20825}$$

$$\therefore \quad \mathrm{a} = 5$$

Q.64 Answer:

$2(\pi + 1)$

Solution:

We have $f(x) - \sin x + \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin x + t f(t)) dt$

Then $f(x)$ is of the form

$= (\pi + 1) \sin x + A$

Where $A = \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} f(t) dt$

Now from A, $A = \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} t((\pi + 1) \sin t + A) dt$

$\Rightarrow A = 2(\pi + 1)$

$\therefore f(x) = (\pi + 1) \sin x + 2(\pi + 1)$

$f_{max} = 3\pi + 3$

$f_{min} = \pi + 1$

Q.65 Answer:

1

Solution:

$x^2 - x + 1 = 0 \Rightarrow x = -\omega, -\omega^2$ and $S = t_1 + t_2 + t_3 + \ldots\ldots\ldots + t_{2011}$

$x = -\omega \Rightarrow t_1 = \left(-\omega + \frac{1}{-\omega}\right)^3 = -\left(\omega + \omega^2\right)^3 = 1, t_2 = \left(\omega^2 + \frac{1}{\omega^2}\right)^3 = \left(\omega^2 + \omega\right)^3 = -1$

$t_3 = \left(-\omega^3 + \frac{1}{-\omega^3}\right)^3 = -(1+1)^3 = -8, t_4 = \left(\omega^4 + \frac{1}{\omega^4}\right)^3 = \left(\omega + \omega^2\right)^3 = -1$

$t_5 = \left(-\omega^5 + \frac{1}{-\omega^5}\right)^3 = -\left(\omega^2 + \omega\right)^3 = 1, t_6 = \left(\omega^6 + \frac{1}{\omega^6}\right)^3 = (1+1)^3 = 8$

$\sum_{r=1}^{6} t_r = 0$. The sum of 6 consecutive terms is 0

$2011 = 6 \times 335 + 1. S = t_{204} = t_1 = 1$

$\sum_{r=1}^{6} t_r = 0$. **The sum of 6 consecutive terms is 0**

$2011 = 6 \times 335 + 1. S = t_{204} = t_1 = 1$

Q.66 Answer:

2

Solution:

$f(f(x)) = \frac{a\left(\frac{ax+b}{bx-a}\right) + b}{b\left(\frac{ax+b}{bx-a}\right) - a} = \frac{a^2x + ab + b^2x - ab}{abx + b^2 - abx + a^2} = \frac{x\left(a^2 + b^2\right)}{\left(a^2 + b^2\right)}$

$f(f(x)) = x$

$g(x) = x + \frac{1}{x} = \frac{5}{2} = 2 + \frac{1}{2}$

$\longrightarrow x = 2$ or $x = 1/2$

Q.67 Answer:

5

Solution:

$A^2 = A$

$A(A - I) = 0 \Rightarrow A = I$

$A^\top = I; \text{adj } A = I$

$\frac{1 - \cos x}{x^2} \to \frac{1}{2}$

$\frac{\tan x - x}{x^3} = \frac{1}{3}$

$\frac{x + x^3/3 - x}{x^3}$

$\frac{e^x - x - 1}{x^2}$

$\frac{e^x - 1}{2x} = \frac{1}{2}$

$$L = \text{Lim}_{x\to 0} \frac{\ln\left(1+\sin^6 x\right)\cdot 2\sin x\cdot\cos x}{\left(e^x-1-x\right)(\tan x - x)(1-\cos x)}$$

$$= \text{Lim}_{x\to 0} \frac{\ln\left(1+\sin^6 x\right)}{\sin^6 x} \cdot \frac{2\sin^7 x\cdot\cos x}{\frac{(1-\cos x)}{x^2}\cdot\frac{(\tan x - x)}{x^3}\frac{(e^x-x-1)}{x^2}} \times \frac{1}{x^2}\times\frac{1}{x^3}\times\frac{1}{x^2}$$

$$= 2\times 2\times 3\times 2 = 24$$

$$|\operatorname{adj} B| = |B|^2 = 1+0+0+24$$

$$|B|^2 = 25 \Rightarrow |B| = 5$$

Q.68 Answer:

All points except two points

Solution:

$$f(x)\begin{cases} 3-x & x<-3 \\ 6 & -3\le x\le 3 \\ 3+x & x>3 \end{cases}$$

Since these expressions are linear function in x or a constant

It is clearly differentiable at all points except at the border points at -3 and 3

At $x=-3, LHD=-1, RHD=0$

At $x=3, LHD=0, RHD=1$

At $x=-3$ **and** $x=3$ **it is not differentiable**

Q.69 Answer:

$\frac{8}{\sqrt{17}+3}$

Solution:

$$\mathrm{I}_2 = (\cos x)^{\sqrt{17}+4}(\sin x)\Big)_0^{\pi/2}$$

$$-\left(\sqrt{17}+4\right)\int(\cos x)^{\sqrt{17}+3}\left(\cos^2 x-1\right)dx$$

$$\Rightarrow I_2 = 0-\left(\sqrt{17}+4\right)\left(I_2-I_1\right)$$

$$\Rightarrow I_2(\sqrt{17}+5) = (\sqrt{17}+4)I_1$$

$$\Rightarrow \frac{I_1}{I_2} = 1+\frac{1}{\sqrt{17}+4} \Rightarrow \frac{I_1}{I_2} = 1+\sqrt{17}-4$$

$$= \sqrt{17}-3$$

Q.70 Answer:

$\left(\frac{2}{3}, -\frac{7}{3}, \frac{5}{3}\right)$

Solution:

Equation of $\frac{x-2}{1} = \frac{y+1}{1} = \frac{3-3}{1} = \lambda$

for point $A, Z=0 \Rightarrow \lambda = -3 \Rightarrow A(-1,-4,0)$

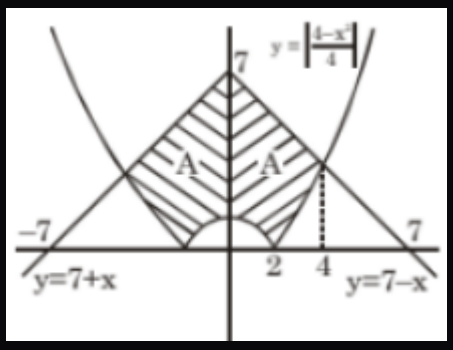
Similarly $B(0,-3,1)$ and $C(3,0,4)$

centroid of $\mathrm{AABC} \equiv \left(\frac{2}{3}, -\frac{7}{3}, \frac{5}{3}\right)$

Q.71 Answer:

—

Solution:



Required Area $= 2$ A

$$A = \int_0^2 \left[7 - x - \left(\frac{4-x^2}{4}\right)\right] dx + \int_2^4 \left[(7-x) - \left(\frac{x^2-4}{4}\right)\right] dx$$

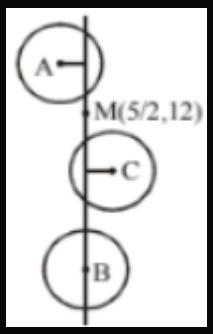
$\Rightarrow A = 16$

$\therefore$ Ans : 32 sq. units

Q.72 Answer:

—

Solution:



$\mathrm{A}(0,16), \mathrm{B}(3,0), \mathrm{C}(5,8)$

line will pass through the mid point of A & C

$\mathrm{m_{MB}} = \frac{12}{-1/2} = -24$

$\left|\frac{m}{8}\right| = 3.$

Q.73 Answer:

—

Solution:

Given

Misplaced \hline

&

Frequency
$(\boldsymbol{f}_{\boldsymbol{i}})$

& \mathbf{x}_{\boldsymbol{i}} & \mathbf{x}_{\boldsymbol{i}} \boldsymbol{f}_{\boldsymbol{i}} & \left(\overline{\boldsymbol{x}}-\mathbf{x}_{\boldsymbol{i}}\right)^2 & \boldsymbol{f}_{\boldsymbol{i}}\left(\overline{\boldsymbol{x}}-\mathbf{x}_{\boldsymbol{i}}\right)^2 \\ \hline 0-10 & 2 & 5 & 10 & 196 & 392 \\ \hline 10-20 & 3 & 15 & 45 & 16 & 48 \\ \hline 20-30 & 4 & 25 & 100 & 36 & 144 \\ \hline 30-40 & 1 & 35 & 35 & 256 & 256 \\ \hline & N=\Sigma f_i=10 & & \Sigma \times f_i=190 & \Sigma f_i\left(\overline{\mathbf{x}}-x_i\right)^2=840 \\ \hline \end{array}

$\because \bar{x}=\frac{\Sigma x_i f_i}{N}=\frac{190}{10}=19$

$\therefore$ Variance $(\sigma)^2=\frac{1}{N} \Sigma f_i\left(\bar{x}-x_i\right)^2=\frac{1}{10}(840)=84$

Q.74 Answer:

—

Solution:

$$
\begin{aligned}
&{}^{2000}C_0+{}^{2000}C_2 3^2+{}^{2000}C_4 3^4 \ldots \ldots \ldots {}^{2000}C_{2009} 3^{2000} \\
&=\left((1+3)^{2000}-(1-3)^{2000}\right) \frac{1}{2} \\
&=\frac{4^{2000}+2^{2000}}{2}=2^{1999}\left(2^{2000}+1\right)
\end{aligned}
$$

Q.75 Answer:

—

Solution:

$\frac{a_{n-1} a_{n+1}-a_n^2}{2 a_{n-1} \cdot a_n}=\frac{1}{n\left(n^2-1\right)} \Rightarrow \frac{a_{n+1}}{a_n}-\frac{a_n}{a_{n-1}}=\frac{2}{n(n+1)(n-1)}=\frac{(n+1)-(n-1)}{n(n+1)(n-1)}$

$$
\begin{aligned}
&\frac{a_{n+1}}{a_n}-\frac{a_n}{a_{n-1}}=\frac{1}{n(n-1)}-\frac{1}{n(n+1)} \\
&\frac{a_n}{a_{n-1}}-\frac{a_{n-1}}{a_{n-2}}=\frac{1}{(n-1)(n-2)}-\frac{1}{(n-1) n} \\
&\frac{a_3}{a_2}-\frac{a_2}{a_1}=\frac{1}{2}-\frac{1}{3}
\end{aligned}
$$

$$\frac{a_{n+1}}{a_n}-\frac{a_2}{a_1}=\frac{1}{2}-\frac{1}{n(n+1)}$$

$$\frac{a_{n+1}}{a_n}=\frac{3}{2}-\left(\frac{1}{n}-\frac{1}{n+1}\right)$$

$$S=\sum_{k=2}^{2000}\frac{a_{k+1}}{a_k}=\sum_{k=2}^{2000}\frac{3}{2}-\left(\frac{1}{k}-\frac{1}{k+1}\right)$$

$$S=\frac{3}{2}\times 1999-\left[\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\ldots+\frac{1}{2000}-\frac{1}{2001}\right]$$

$$=\frac{3\times 1999}{2}-\frac{1}{2}+\frac{1}{2001}$$

$$=\frac{5997-1}{2}+\frac{1}{2001}$$

$$S=2998+\frac{1}{2001}$$

$$[S]=2998$$